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Optimizing gift card and pricing strategies in the presence of double mental discounting

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ABSTRACT

Gift card promotions are widely used in the retail industry and often lead to a phenomenon known as double mental discounting, where consumers mentally deduct the gift card value from their purchase costs multiple times—both when they receive and redeem the card. This paper develops an analytical model to examine the impact of double mental discounting on retailers' promotion and pricing decisions within a two-period framework. The results indicate that retailers' decisions to implement gift card strategies depend on both the double mental discounting and the cash equivalent effects. When offering gift card promotions, retailers generally provide high-value gift cards and raise prices to stimulate consumer purchases while preserving revenue. The double mental discounting effect positively influences gift card value but may negatively impact prices. Additionally, when consumers exhibit inconsistency in their perceived value of gift cards, retailers are even more inclined to offer gift card promotions. Furthermore, we extend the basic model to incorporate consumer heterogeneity in cash equivalent effects, retailer sales costs and competition. This study provides valuable managerial insights for retailers, particularly in identifying when and how to deploy gift card promotions in the presence of double mental discounting.

1. Introduction

Gift card promotions offer customers incentives by providing store-specific gift cards when they purchase selected products or when their total purchase amount reaches a certain threshold (Khouja et al., 2011, 2013a). Retailers in the electronics, sports, and luxury industries have increasingly adopted gift cards as a key promotional tool (Chen et al., 2022; Cheng and Cryder, 2018; Kadiyala et al., 2024). For instance, as one of the world's largest consumer electronics companies, Apple has utilized gift card promotions during major shopping events such as Black Friday and the Education Savings program.[1,2] Specifically, in the 2023 Black Friday event, Apple offered up to $200 in free gift cards with the purchase of select products. Similarly, High Wave Sports of Wanxiang Hui in China provides "Nike's Cash" to customers, where buyers of $200 Nike Air Max shoes receive a $20 Nike gift card for future purchases (Chen et al., 2022). Additional examples of gift card promotions in the electronics, sports, and luxury sectors are detailed in Appendix A.

Gift card promotions influence purchase decisions across two stages. First, customers receive gift cards when purchasing selected products, significantly impacting the initial purchase decision. Second, the value of the gift card is applied to the subsequent purchase, directly reducing the cost of that purchase. The nature of gift cards, being tied to multiple purchases, often leads to the phenomenon of double mental discounting. *Double mental discounting* refers that when a single gain (e.g., a gift card) is associated with multiple purchases, consumers tend to mentally subtract this gain from perceived costs multiple times (Cheng and Cryder, 2018). For example, consider Apple's "Education Savings" promotion illustrated in Fig. 1. Suppose a student purchases an Apple iPad Air for $549 and receives a $100 gift card for future use. The satisfaction of receiving the gift card may lead the student to mentally offset the iPad's cost, thinking, "I'm really spending only $449 on this iPad because I'm getting $100 back." A few months later, the student returns to the Apple store and redeems the $100 gift card toward a $399 Apple Watch, effectively reducing the cost to $299, prompting him to think, "I'm spending only $299 on this watch thanks to the gift card."

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[1] Apple's gift card in Black Friday https://www.macrumors.com/2023/11/16/apple-black-friday-shopping-event-gift-cards/.

[2] Apple's gift card in Education saving https://www.apple.com/us-edu/store.

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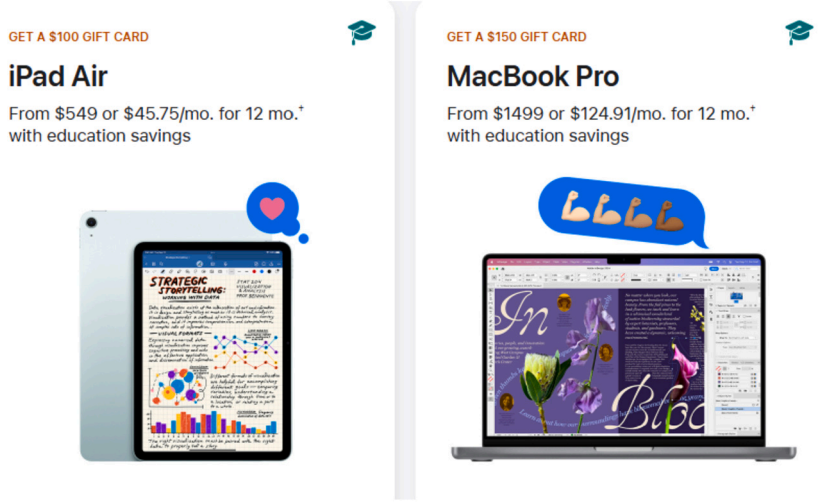


Fig. 1. Gift card promotions in Apple.

Double mental discounting differs from traditional discounting in two key aspects. First, they vary in the frequency of mental deductions. With the double mental discounting, consumers mentally deduct the discount multiple times. Whereas with the traditional discounting, consumers subtract the discount only once. Second, they have distinct impacts on consumer spending. Double mental discounting often leads consumers to spend more, as they perceive a lower price across multiple periods, while traditional discounting may not generate the same effect (Cheng and Cryder, 2018). For retailers, double mental discounting leads consumers to perceive lower prices, presenting an opportunity to leverage this behavioral bias through strategic pricing adjustments. However, failing to account for this bias could expose retailers to substantial potential losses. In previous literature, customers are typically assumed to be fully rational, where the value of a gift card is deducted only once during the redemption period to represent the mental accounting process (Khouja et al., 2013a; Khouja and Zhou, 2010). Therefore, examining the influence of double mental discounting on retailers' gift card strategies is of particular value for both practice and academic.

When offering gift cards, retailers must consider two key factors. First, consumers do not view gift cards as fully equivalent to cash (Khouja et al., 2013a,b). Evidence from secondary markets shows that consumers value each dollar of gift card at roughly $0.75 to $0.90 in cash terms (Offenberg, 2007). This phenomenon, termed the *cash equivalent effect* in this paper, captures consumers' mental discounting of the gift card's face value. Second, this cash equivalent effect can vary in consistency. Given the time gap between the initial purchase and the eventual redemption of the gift card, consumers face two separate decision points: the purchase decision and the redemption decision (Khouja et al., 2011). These decisions may be either consistent or inconsistent (Jain, 2019; Khouja et al., 2013a). For consistent consumers, the cash equivalent effect remains the same when they receive and redeem the gift card. In contrast, inconsistent consumers experience a shift in their cash equivalent effect at redemption, differing from their initial perception at the time of purchase.

Considering double mental discounting, the cash equivalent effect, and consumer inconsistency, this paper addresses two main questions: First, how do double mental discounting and the cash equivalent effect influence a retailer's pricing and gift card strategies? Specifically, under what conditions does the retailer implement gift card promotions, and how should they design these pricing and gift card strategies? Second, what is the effect of consumer inconsistency on the retailer's pricing and promotional strategies? In particular, how do the retailer's strategies and profitability change when consumers are inconsistent versus consistent in their behavior?

This paper develops a two-period decision model to investigate the optimal pricing and gift card strategies for retailers, considering double mental discounting and the cash equivalent effect. We begin by examining consumer behavior under consistent preferences and then analyze scenarios involving consumer inconsistency. Our findings indicate that a retailer's decision to implement gift card strategies is influenced by both the double mental discounting and the cash equivalent effects. Specifically, when the cash equivalent effect is low, the optimal strategy is to avoid gift card promotions; otherwise, offering gift cards becomes profitable. In cases where gift cards are offered, retailers may select from one of three distinct strategies, with the choice closely linked to the relative strength of these effects. When offering gift card promotions, retailers typically provide high-value gift cards and raise prices to boost purchases and maintain revenue. The double mental discounting effect positively influences gift card value but may negatively affect prices. Additionally, when consumers exhibit inconsistency, retailers are more likely to use gift card promotions, leading to increased retailer profit and consumer surplus, as supported by numerical results. Numerical experiments with exponential and normal distributions confirm that the insights derived from a uniform distribution remain robust. Finally, we extend the basic model to account for consumer heterogeneity in cash equivalent effects, retailer sales costs, and market competition, and derive firms' optimal strategies.

The paper is structured as follows: Section 2 reviews the relevant literature. Section 3 introduces the model and presents the optimal decisions for consumers with consistent behavior, while Section 4 examines consumer inconsistency. Section 5 offers a numerical analysis, and Section 6 extends the model to account for consumer heterogeneity in the cash equivalent effect, retailer sales costs, and market competition. Finally, Section 7 concludes the paper. All proofs are provided in Appendix B.

2. Literature review

Our research builds on three key areas: the impact and strategic use of gift cards, principles of mental accounting, and the influence of behavioral biases on pricing and revenue management.

2.1. Gift cards

Numerous studies have empirically explored how gift cards influence retailer profits and consumer behavior (Carlson and Paul, 2022; Kadiyala et al., 2024; Norvell and Horky, 2017). For example, Norvell and Horky (2017) demonstrates that while gift cards can indeed boost a firm's revenue,

they may also have adverse effects on overall profitability. Similarly, Kadiyala et al. (2024) shows that gift cards can significantly enhance both the likelihood of consumer purchases and total spending. However, when consumers assess the value of gift cards, they typically perceive them as less valuable than an equivalent amount of cash (Offenberg, 2007). This reduced valuation arises for two main reasons: first, gift cards are only redeemable at the issuing retailer, and second, there is a chance consumers might not redeem the card at all. Together, these factors contribute to what is known as the cash equivalent effect, where gift cards are seen as worth less than their nominal cash value. Additionally, Gunasti and Baskin (2018) explores consumer preferences and valuations between luxury and non-luxury brand gift cards, finding that luxury brand gift cards are valued lower than those from non-luxury brands.

Considering the cash equivalent effect, prior research has analytically examined how gift card promotions influence customer purchasing behavior and the conditions under which they benefit retailers (Cao et al., 2020; Khouja et al., 2013a). Khouja et al. (2011) investigate firms' optimal gift card strategies and find that, without spending constraints, the optimal strategy is to provide at most one gift card, whereas with spending constraints, offering multiple gift cards becomes optimal. Additionally, Zhang et al. (2018) examine the influence of gift cards on retailers' ordering decisions, showing that gift cards can stimulate demand and increase profits. Research on gift cards also extends to the supply chain context. For instance, Khouja et al. (2013b) explore manufacturer-sponsored gift cards and find that these can enhance supply chain efficiency. Li et al. (2021) expand the analysis to include two-product scenarios, comparing cases without gift cards, manufacturer-sponsored gift cards, and retailer-provided gift cards, concluding that retailer-sponsored gift cards are optimal in a two-product supply chain. Similarly, Li et al. (2022b) investigate pricing strategies for retailers and manufacturers, finding that the provision of gift cards leads to higher retail and wholesale prices compared to scenarios without gift cards.

Previous research predominantly addresses the cash equivalent effect, often treating gift cards as one-time deductible promotions and giving limited attention to consumers' double mental discounting behavior. To our knowledge, only Chen et al. (2022) examine double mental discounting in the context of gift card promotions. Their study explores how a supplier and retailer strategically use instant discounts and gift card promotions while accounting for this behavioral bias. Our work differs from theirs in two key ways. First, while their model assumes that all consumers exhibit double mental discounting, evidence shows that consumer responses to marketing tactics are heterogeneous (Geng and Shulman, 2015). Our paper therefore introduces consumer heterogeneity, assuming only a subset of consumers exhibit double mental discounting. Second, while their model treats product prices as exogenous, we incorporate the retailer's strategic pricing decisions. Thus, this paper advances the field by integrating consumer heterogeneity in double mental discounting and including retailers' strategic pricing decisions. Additionally, our study further contributes by examining the combined influence of both the cash equivalent effect and double mental discounting on retailers' decision-making.

2.2. Mental accounting

Mental accounting describes how consumers mentally organize, budget, and evaluate their financial activities (Thaler, 1985). It has been used to explain various consumption and spending behaviors (Erat and Bhaskaran, 2012; Lim et al., 2024). Research has demonstrated that mental accounting can lead to consumption biases, including tendencies toward under- or over-consumption (Banerjee et al., 2019; Thaler, 1999). For example, Ülkü et al. (2020) find that consumers, influenced by mental accounting for sunk costs, tend to increase consumption by amortizing fixed (or sunk) costs, such as time spent in a queue. Additionally, Banerjee et al. (2019) show that consumption biases driven by mental accounting vary by cultural context: Eastern consumers with holistic thinking styles are less affected by these biases, while Western consumers with analytical thinking styles are more prone to them. Other studies examine how mental accounting shapes consumer behavior with windfalls. For instance, Milkman and Beshears (2009) found that receiving a windfall increased spending on online groceries, while Liu et al. (2021) observe that consumers treat digital coupons as windfalls, leading to a higher marginal propensity to consume.

The role of mental accounting in gift card promotions and its influence on consumer purchasing decisions has also been widely studied. Helion and Gilovich (2014) suggest that people tend to allocate the value of a gift card to a hedonic mental account, leading them to spend it on more indulgent purchases. Similarly, Reinholtz et al. (2015) propose that once a mental account is established, consumers are more likely to choose purchases that align with that account, favoring options that are congruent with the mental budget. Through experiments, they demonstrate that consumers prefer retailer-specific items when using retailer-specific gift cards. Additionally, Cheng and Cryder (2018) extend the concept of malleable mental accounting by proposing that, in gift card promotions, consumers apply the value of a single gift card to multiple expenses, thereby lowering their perceived costs—a behavior known as "double mental discounting".

These empirical studies support our theoretical analysis, demonstrating the influence of mental accounting on consumer purchasing behavior and highlighting how double mental discounting, as driven by mental accounting, increases consumer spending. Building on mental accounting theory, this paper models the decision-making process underpinning double mental discounting in gift card promotions. By quantifying this bias, we examine its effects on firms' strategic pricing and promotion decisions, offering new insights into how mental accounting shapes consumer responses to promotional tactics.

2.3. Behavioral biases

Behavioral biases in pricing and revenue management represent a significant area of research in operations management. Loss aversion, a core concept from prospect theory, posits that consumers perceive losses more intensely than equivalent gains (Kai-Ineman and Tversky, 1979; Karle et al., 2015). One stream of recent research has extensively examined the influence of decision-makers' loss aversion on supply chains using analytical models (Li et al., 2022a, 2023; Xie et al., 2024). Another stream has explored how firms strategically respond to consumer loss aversion (Das and Dutta, 2022; Zhang and Li, 2021). For instance, Yang et al. (2018) investigated a service system where customers exhibit loss aversion toward both price and delay, and derive an optimal pricing strategy for the monopoly provider. Chen and Nasiry (2020) explored how a firm's pricing strategies are influenced by loss-averse consumers with heterogeneous perceptions of gains and losses in the price dimension. Aligned with Yang et al. (2018) and Chen and Nasiry (2020), this paper examines the impact of consumers' double mental discounting on retailers' pricing and promotion strategies, offering new insights into how behavioral biases influence decision-making in retail contexts.

Framing effect, another well-known behavioral bias, refers to how different presentations of the same decision problem can lead to cognitive biases in individuals, which, in turn, influence their decision-making behavior (Tversky and Kahneman, 1981). This effect can be either positive or negative. Positive framing highlights the potential gains or benefits of an action or decision, whereas negative framing emphasizes potential losses or adverse outcomes (Florence et al., 2022). Experimental studies have extensively explored the framing effect across various decision-making contexts,

Table 1
Symbol description.

Symbol	Description
v	Product valuation
p_i^j	Product price ($i = 1, 2$, and $j = N, 1, 2, 3, 4$)
G^j	Gift card face value ($j = 1, 2, 3, 4$)
γ_i	Cash equivalent effect ($i = 1, 2, H, L$)
α	Double mental discounting effect
β	Proportion of consumers with high cash equivalent effect
c	Unit sales cost
t	Degree of horizontal differentiation
D_{ik}	Consumer's demand ($i = 1, 2$, and $k = m, n$)
cs^j	Consumer surplus ($j = N, 1, 2, 3$)
π^j	Retailer's profit ($j = N, 1, 2, 3, 4$)

such as crowdfunding (Kuo et al., 2022), food waste (Dai and Gong, 2024; Zheng et al., 2023), healthcare (Liu and Kc, 2023), and pricing strategies (Alnes and Haugom, 2024; Koo and Suk, 2020). However, fewer studies have analyzed the framing effect using analytical models (Ertekin et al., 2019; Moon and Shugan, 2018). Ertekin et al. (2019) investigated how consumer behavioral responses to a retailer's discount framing influence retail sales and return performance. Their model shows that consumers' perceived prices systematically deviate from actual prices when stacked discounts are presented. Similar with Ertekin et al. (2019), we extend the analysis to examine scenarios where consumers perceive prices to be lower than their actual values in the context of gift card promotions.

Double mental discounting shares similarities with loss aversion and the framing effect, but also has unique characteristics. All three biases stem from prospect theory and cause deviations from the predictions of expected utility theory in consumer decision-making (Kim et al., 2023; Zhang and Li, 2021; Zheng et al., 2023). Double mental discounting reduces consumers' perceived costs, thereby encouraging greater spending (Cheng and Cryder, 2018). In contrast, loss aversion emphasizes the role of a reference point in determining whether an outcome is perceived as a loss or a gain (Xie et al., 2024). Meanwhile, the framing effect suggests that individual preferences are influenced by how choices are presented (Lei et al., 2022). Although extensive research covers loss aversion and framing effect, double mental discounting remains less explored. This paper aims to address this gap by analyzing the effects of double mental discounting on retailers' strategic choices, offering new insights to deepen both academic understanding and practical applications in marketing and pricing strategies.

3. Model framework and analysis

This paper considers a retailer engaging in a gift card promotion over two periods. The first period involves the promotion of the gift card, while the second period is when the gift card can be redeemed. Upon purchasing a product in the first period, consumers receive a gift card with a face value of G, which can be redeemed in the second period. The product prices are p_1 in the first period and p_2 in the second period. Following Yan et al. (2019) and Pei et al. (2021), we assume zero sales costs to simplify the computations and maintain model tractability. In Section 6, we extend our model to incorporate retailers' sales costs. The retailer's objective is to maximize profit.

Consumers have a unit demand in each period and can purchase at most one product per period (Chen et al., 2022; Jalili and Pangburn, 2021). The value consumers assign to the product, denoted as v, follows a uniform distribution within the interval $[0, 1]$ (Yan et al., 2016; Zhou et al., 2024). Since gift cards can only be redeemed at specific retailers or may not be redeemed due to loss or other reasons, consumers do not consider gift cards to be equivalent to cash (Khouja et al., 2013a). Therefore, we assume that the cash equivalent value of a \$1 gift card is γ_1 in the first period and γ_2 in the second period ($0 < \gamma_i < 1$, $i = 1, 2$) (Chen et al., 2022). It is well established that consumer segments vary in their responsiveness to marketing strategies (Geng and Shulman, 2015). Thus, this paper assumes that there are two distinct segments: a proportion α of consumers exhibit double mental discounting, while the remaining proportion, $1 - \alpha$, does not. This assumption is similar to that of Fang et al. (2021), who assume that consumers are heterogeneous in terms of their disutility from order frequency. Hereafter, we refer to α as the double mental discounting effect. The potential market size is standardized to 1. Consumer decision-making is based on utility maximization. Since the time between the purchase and redemption periods is short, we do not consider the time value of money (Khouja et al., 2011).

This paper assumes that the retailer first determines the product price and the face value of the gift card, after which consumers make their purchase decisions. It is further assumed that information is symmetrical between the retailer and consumers. Table 1 presents the main symbols used throughout the analysis. Subscripts m and n denote consumers with and without double mental discounting, respectively. Subscripts 1 and 2 correspond to the first and second periods, respectively. Superscripts $N, 1, 2, 3$ represent scenarios without gift cards and with gift card strategies 1, 2, and 3, respectively. The subscript I indicates consumer inconsistency.

3.1. Consumers purchase behavior

We consider that there are two types of consumers: consumers with double mental discounting and consumers without double mental discounting. Consumers with double mental discounting deduct the gift card's value from their perceived purchase cost multiple times, while those without this effect apply the deduction only at the redemption stage. We assume that the cash equivalent effect when consumers make their redemption decisions is the same as at the time of their initial purchase, that is $\gamma_1 = \gamma_2 = \gamma$ and $0 < \gamma < 1$.

(1) Consumers with double mental discounting

In the first period, consumers with double mental discounting will make a purchase and receive the gift card if $v - (p_1 - \gamma G) \geq 0$; otherwise, they will not purchase. Consequently, the demand in the first period is given by $D_{1m} = 1 - (p_1 - \gamma G)$. In the second period, consumers with a gift card will purchase if $v - (p_2 - \gamma G) \geq 0$. If $v - (p_1 - \gamma G) < 0$ but $v - p_2 \geq 0$, consumers without a gift card will purchase. Therefore, the demand in the second period is $D_{2m} = 1 - \max\{p_1 - \gamma G, p_2 - \gamma G\} + \max\{p_1 - \gamma G - p_2, 0\}$.

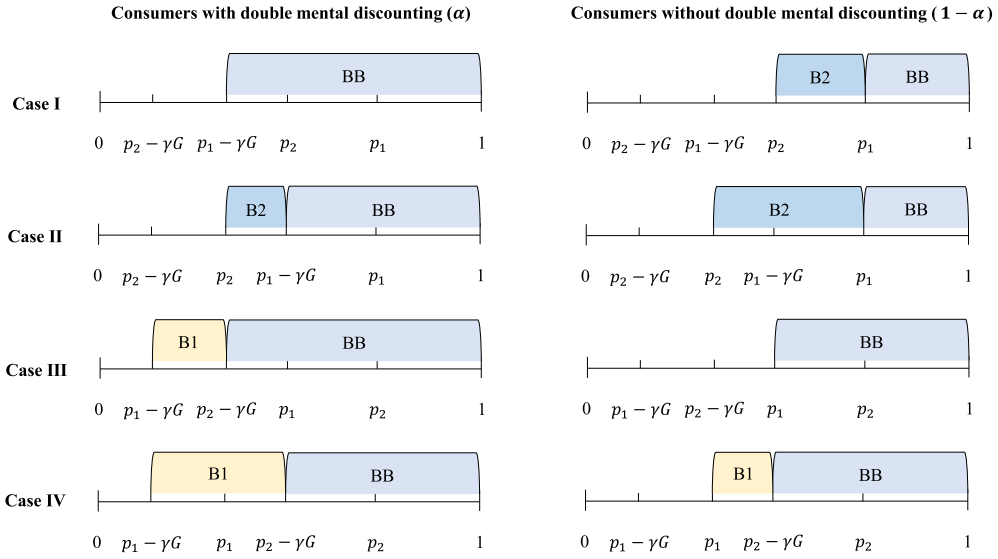


Fig. 2. Consumers' purchase decisions.

(2) Consumers without double mental discounting

Gift cards directly reduce the purchase cost during redemption, so consumers without double mental discounting only deduct the gift card's value at the redemption stage. In the first period, consumers without double mental discounting will purchase and receive the gift card if $v - p_1 \geq 0$; otherwise, they will not purchase. Thus, the demand in the first period is $D_{1n} = 1 - p_1$. In the second period, consumers with a gift card will purchase if $v - (p_2 - \gamma G) \geq 0$. If $v - p_1 < 0$ but $v - p_2 \geq 0$, consumers without a gift card will purchase. Therefore, the demand in the second period is $D_{2n} = 1 - \max\{p_1, p_2 - \gamma G\} + \max\{p_1 - p_2, 0\}$.

Fig. 2 summarizes consumers' purchase decisions, which are categorized into four cases. Specifically, "BB" denotes purchasing in both periods, "B1" indicates purchasing only in the first period, and "B2" represents purchasing only in the second period.

3.2. Equilibrium analysis

Based on the above analysis, the retailer's profit functions under four cases are as follow,

Case I

$$\max_{p_1,p_2,G} \pi = \underbrace{p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from product sales in period 1}} + \underbrace{p_2(1-\alpha)\int_{p_2}^{p_1} 1\,dx}_{\text{Profit from consumers without gift cards in period 2}} + \underbrace{(p_2-G)\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from consumers with gift cards in period 2}}, \quad (1)$$

$$\text{subject to} \quad 1 \geq p_1 \geq p_2 > p_1 - \gamma G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0.$$

Case II

$$\max_{p_1,p_2,G} \pi = \underbrace{p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from product sales in period 1}} + \underbrace{p_2\left(\alpha\int_{p_2}^{p_1-\gamma G} 1\,dx + (1-\alpha)\int_{p_2}^{p_1} 1\,dx\right)}_{\text{Profit from consumers without gift cards in period 2}} + \underbrace{(p_2-G)\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from consumers with gift cards in period 2}}, \quad (2)$$

$$\text{subject to} \quad 1 \geq p_1 > p_1 - \gamma G \geq p_2 \geq 0 \quad \text{and} \quad 1 \geq G \geq 0.$$

Case III

$$\max_{p_1,p_2,G} \pi = \underbrace{p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from product sales in period 1}} + \underbrace{(p_2-G)\left(\alpha\int_{p_2-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from consumers with gift cards in period 2}}, \quad (3)$$

$$\text{subject to} \quad 1 \geq p_2 \geq p_1 > p_2 - \gamma G > p_1 - \gamma G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0.$$

Case IV

$$\max_{p_1,p_2,G} \pi = \underbrace{p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right)}_{\text{Profit from product sales in period 1}} + \underbrace{(p_2-G)\int_{p_2-\gamma G}^{1} 1\,dx}_{\text{Profit from consumers with gift cards in period 2}}, \quad (4)$$

$$\text{subject to} \quad 1 \geq p_2 > p_2 - \gamma G \geq p_1 > p_1 - \gamma G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0.$$

Solving equation (1)–(4), we derive Proposition 1, with the proof detailed in Appendix B.1. The expressions for $\hat{\gamma}$, $\tilde{\gamma}$, and $\bar{\gamma}$ are included in the proof of Proposition 1.

Proposition 1. *The optimal strategies for the retailer are,*

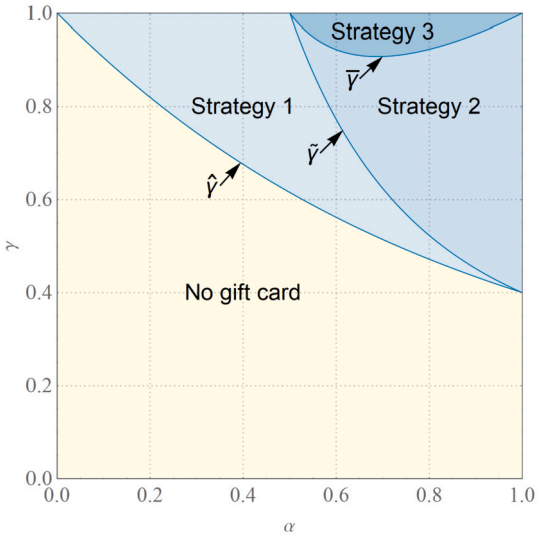


Fig. 3. Retailer's optimal strategies.

(1) when $\gamma \le \hat{\gamma}$, the retailer does not engage in gift card promotions, setting $p_1^{N*} = \frac{1}{2}$ and $p_2^{N*} = \frac{1}{2}$. (No gift card)

(2) when $\alpha \le \frac{1}{2}$ and $\gamma > \hat{\gamma}$, or $\alpha > \frac{1}{2}$ and $\hat{\gamma} < \gamma \le \tilde{\gamma}$, the retailer offers gift cards with $p_1^{1*} = \frac{2+\alpha\gamma}{2(1+\gamma+\alpha\gamma)}$, $p_2^{1*} = 1$, and $G^{1*} = \frac{2+\alpha}{2(1+\gamma+\alpha\gamma)}$. (Strategy 1)

(3) when $\alpha > \frac{1}{2}$ and $\tilde{\gamma} < \gamma \le \bar{\gamma}$, the retailer sets $p_1^{2*} = \frac{(1-\alpha)^2-\alpha^2\gamma(2+\gamma)}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$, $p_2^{2*} = 1$, and $G^{2*} = \frac{2+\alpha^2(1-\gamma)-\alpha(3+2\gamma)}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$. (Strategy 2)

(4) when $\alpha > \frac{1}{2}$ and $\gamma > \bar{\gamma}$, the retailer sets $p_1^{3*} = \frac{1-\alpha-2\alpha\gamma}{2(1-2\alpha-\gamma+\alpha\gamma)}$, $p_2^{3*} = 1$, and $G^{3*} = \frac{1-\alpha-2\alpha\gamma}{2\gamma(1-2\alpha-\gamma+\alpha\gamma)}$. (Strategy 3)

Proposition 1 and Fig. 3 indicate that the decision to implement a gift card promotion depends on the proportion of consumers exhibiting double mental discounting (α) and the strength of the cash equivalent effect (γ). When the cash equivalent effect is low ($\gamma \le \hat{\gamma}$), the retailer's optimal strategy is to avoid gift card promotions. However, if the cash equivalent effect is high ($\gamma > \hat{\gamma}$), offering gift cards becomes profitable. Specifically, when the proportion of consumers with double mental discounting is relatively low, the retailer adopts Strategy 1. As this proportion increases, the retailer shifts to Strategy 2. Finally, when both the proportion of double mental discounting consumers and the cash equivalent effect reach sufficiently high levels, the retailer employs Strategy 3.

A key managerial implication of Proposition 1 and Fig. 3 is that both the proportion of consumers with double mental discounting and the cash equivalent effect are critical in guiding retailers on whether and how to implement gift card promotions. First, retailers should assess the proportion of consumers exhibiting double mental discounting, as a higher proportion leads to increased consumer overspending, which benefits retailers. They must also take into account the cash equivalent effect, which is influenced by objective factors like store accessibility and product variety (Gunasti and Baskin, 2018). By optimizing these factors, retailers can enhance the cash equivalent effect and improve the effectiveness of gift card promotions. When either the cash equivalent effect or the proportion of consumers with double mental discounting is low, retailers should avoid offering gift cards. However, when both are high, implementing gift card promotions becomes advantageous. Finally, retailers should adjust their pricing and gift card strategies based on the proportion of consumers with double mental discounting and the strength of the cash equivalent effect.

According to Proposition 1, we derive Corollary 1.

Corollary 1. *(1) The product prices with gift cards are higher than without gift cards ($p_i^{j*} > p_i^{N*}$, $i = 1,2$ and $j = 1,2,3$). (2) When offering a gift card promotion, the gift card value is higher than the first-period price ($G^{j*} > p_1^{j*}$, $j = 1,2,3$).*

Corollary 1 suggests that, due to the redemption cost associated with gift cards, retailers tend to raise product prices when implementing such promotions. This finding is consistent with previous research, where Khouja et al. (2013b) and Li et al. (2022b) also noted that retailers increase prices during gift card promotions. Additionally, retailers often set the gift card value significantly higher to attract customers. For example, as detailed in Appendix A, Apple offered up to $200 in free gift cards during the 2023 Black Friday sales, and Beaverbrooks provided a free e-gift card worth up to £350.

According to our model, retailers should increase prices and offer high-value gift cards when running gift card promotions. Double mental discounting causes consumers to perceive the price as lower than it actually is. Furthermore, offering high-value gift cards further reduces the perceived price. Thus, this strategy can effectively help retailers boost sales and increase revenue by raising prices while offering attractive gift cards.

Building on Proposition 1, we drive consumer surplus calculations in Appendix B.3. By comparing consumer surplus with and without gift card promotions, we obtain Corollary 2.

Corollary 2. *Consumer surplus with gift card promotions can be higher than without gift card promotions.*

According to Corollary 1, the retailer provides a high-value gift card that can cover a substantial portion of the purchase cost in the second period. At the same time, implementing a gift card promotion leads the retailer to raise product prices in both periods, which increases the consumers' purchase cost. Whether consumer surplus is higher with gift card promotions depends on the balance between these effects: if the gain from the gift card exceeds the increased costs, consumer surplus will be higher; otherwise, it will be lower.

Cheng and Cryder (2018) find that, in the context of gift cards, consumers exhibit an intriguing duality when differentiating between perceived and actual costs. First, double mental discounting lowers consumers' perceived purchase cost, which can increase the likelihood of overspending. However, consumers also tend to remain aware of the actual costs and are not easily "tricked" by gift card promotions. This duality leads to a nuanced impact of gift cards on consumer surplus, aligning with Corollary 2. Therefore, our results further support the empirical findings of Cheng and Cryder (2018), demonstrating that consumer surplus can indeed be higher with gift card promotions.

3.3. The impact of double mental discounting and cash equivalent effects

In this subsection, we examine the impacts of double mental discounting and cash equivalent effects on prices and gift card values. Proofs are provided in Appendix B.4.

Corollary 3. *The price p_1^{1*} decreases in double mental discounting effect, while p_1^{2*} and p_1^{3*} increase as double mental discounting effect intensifies.*

Corollary 3 indicates that the proportion of consumers exhibiting double mental discounting affects first-period pricing across different strategies. Consumers without double mental discounting only consider the gift card's value at the redemption stage, while those with double mental discounting mentally apply the gift card value in both periods, increasing their likelihood of making a purchase in the first period. As illustrated in Fig. 3, Strategy 1 involves a relatively small proportion of consumers with double mental discounting, resulting in a lower probability of first-period purchases. As this proportion grows, gift card promotions continue to have limited impact on first-period decisions, leading the retailer to lower prices to encourage purchases. In contrast, in Strategies 2 and 3, where the proportion of consumers with double mental discounting is relatively high, the probability of first-period purchases rises. To maximize profit from this heightened demand, the retailer responds by increasing prices as the proportion becomes larger.

Corollary 4. *The gift card value G^{j*} $(j = 1, 2, 3)$ increases in double mental discounting effect.*

Corollary 4 indicates that a higher proportion of consumers with double mental discounting leads to a higher gift card value. Gift cards have dual impacts on the retailer's revenue: they encourage consumer purchases, boosting profits, but they also incur redemption costs, which reduce profits. As the proportion of consumers with double mental discounting increases, the likelihood of first-period purchases rises, and a higher gift card value further incentivizes these consumers. Since the additional gains from setting a higher gift card value outweigh the redemption costs, retailers are motivated to increase the gift card value.

Corollary 5. *The price p_1^{1*} decreases in cash equivalent effect, p_1^{2*} and p_1^{3*} increase in cash equivalent effect.*

Corollary 5 highlights that the cash equivalent effect impacts pricing similarly to the proportion of consumers with double mental discounting. For consumers without double mental discounting, an increase in the cash equivalent effect has no impact on first-period purchases. However, for consumers exhibiting double mental discounting, a stronger cash equivalent effect raises the likelihood of first-period purchases. In Strategy 1, as shown in Fig. 3, the double mental discounting effect is limited, so only a small portion of consumers is influenced by gift cards to purchase in the first period. As the cash equivalent effect strengthens, gift card promotions still have limited influence on first-period purchases, encouraging the retailer to lower prices to boost demand. In Strategies 2 and 3, where the double mental discounting effect is stronger, a larger proportion of consumers is incentivized by gift cards to make first-period purchases. As the cash equivalent effect continues to strengthen, gift cards further increase first-period purchases. Meanwhile, the improved cash equivalent effect also raises redemption rates in the second period, increasing retailer costs and prompting higher prices to sustain revenue.

Corollaries 3 and 5 indicate that when implementing gift card promotions, retailers must carefully assess both the proportion of consumers with double mental discounting and the strength of the cash equivalent effect. Corollary 3 reveals that a higher proportion of consumers with double mental discounting does not always lead to higher prices. When this proportion is initially low and starts to rise, retailers should reduce product prices; however, as it reaches higher levels, retailers can leverage it by increasing prices. Similarly, Corollary 5 suggests that when the cash equivalent effect is low and begins to strengthen, retailers should lower prices. Once the cash equivalent effect is already high and continues to grow, raising prices becomes optimal.

Corollary 6. *The gift card value G^{1*} decreases in cash equivalent effect, while G^{2*} increases in cash equivalent effect, and G^{3*} initially increases and then decreases in cash equivalent effect.*

As previously discussed, gift cards have two opposing effects on retailer revenue: they encourage purchases but also raise redemption costs. In Strategy 1, where the proportion of consumers with double mental discounting is relatively low, gift cards only stimulate a small number of purchases. As the cash equivalent effect increases, the impact of gift cards on purchases remains limited, while redemption costs rise significantly, prompting the retailer to lower the gift card value (G^{1*}) to protect profits. In Strategy 2, where the proportion of consumers with double mental discounting is relatively high, gift cards successfully drive most consumers to make purchases. An increase in the cash equivalent effect further stimulates purchases, with the benefits of offering gift cards outweighing the costs, leading the retailer to increase the gift card value (G^{2*}) to incentivize more purchases. In Strategy 3, where both the proportion of consumers with double mental discounting and the cash equivalent effect are high, an initial rise in the cash equivalent effect results in purchase-driven benefits that exceed the costs, motivating the retailer to raise the gift card value (G^{3*}). However, as costs eventually surpass the benefits, the retailer subsequently reduces the gift card value.

4. Consumers inconsistency

Due to the time gap between the initial purchase and the redemption of a gift card, consumers face two distinct decision points: the purchase decision and the redemption decision (Khouja et al., 2013b). These decisions may either be consistent or inconsistent (Jain, 2019; Khouja et al.,

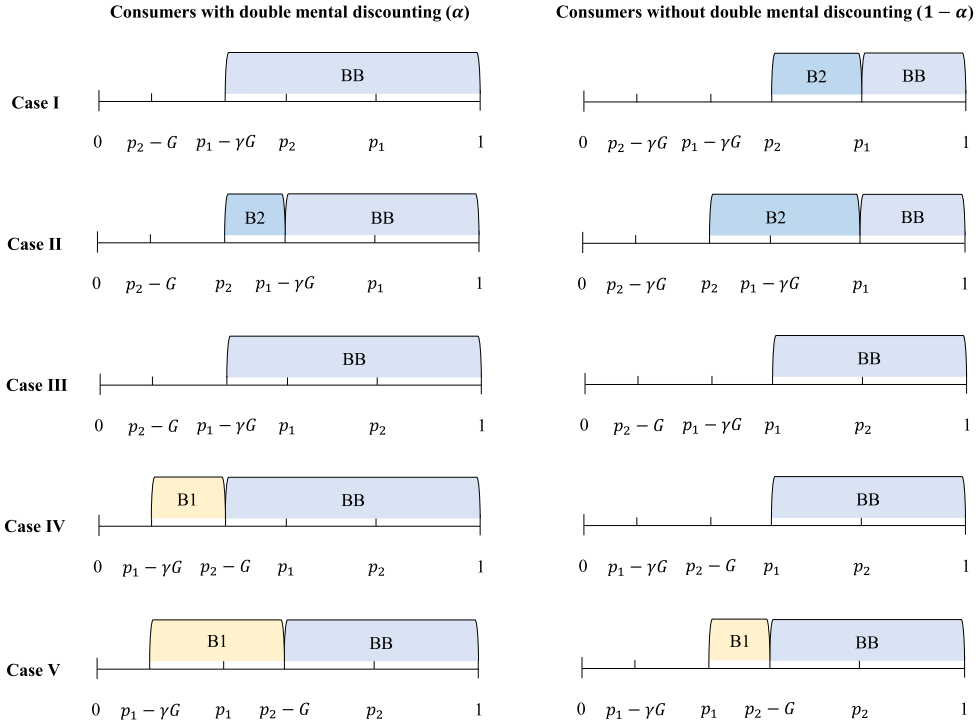


Fig. 4. Consumers' purchase decisions with consumers inconsistency.

2013a). In Section 3, we assumed consumers consistency in the cash equivalent effect when receiving and redeeming the gift card. In this section, we explore consumers inconsistency, which refers to the difference between the expected cash equivalent effect at the time of purchase and the actual effect during gift card redemption (Khouja et al., 2011). When consumers redeem the gift card, its face value partially offsets the purchase cost in the second period, directly reducing the consumer's payment. Thus, it is assumed that the cash equivalent effect during the redemption period is higher than during the initial purchase period ($\gamma_2 > \gamma_1$). Previous research has made similar assumptions to explore consumer inconsistency in contexts like cash mail-in rebates and gift card promotions (Khouja et al., 2013a; Khouja and Zhou, 2010). For simplification, it is assumed that the cash equivalent effect in the first period is $\gamma_1 = \gamma$, while in the second period, it is $\gamma_2 = 1$.

4.1. Consumers purchase behavior

In the first period, consumers' purchase behavior is the same as discussed in Section 3, and we won't repeat the details here. The changes in consumers' behavior occur in the second period. For consumers with double mental discounting, they will purchase in the second period if $v-(p_2-G)\geq 0$; otherwise, they will not purchase. If $v-(p_1-\gamma G)<0$ and $v-p_2\geq 0$, consumers without a gift card will purchase; otherwise, they will not purchase. Therefore, the demand in the second period is $D_{2m}=1-\max\{p_1-\gamma G, p_2-G\}+\max\{p_1-\gamma G-p_2,0\}$. For consumers without double mental discounting, they will purchase if $v-(p_2-G)\geq 0$; otherwise, they will not purchase. If $v-p_1<0$ and $v-p_2\geq 0$, consumers without a gift card will purchase; otherwise, they will not purchase. Hence, the demand in the second period is $D_{2n}=1-\max\{p_1,p_2-G\}+\max\{p_1-p_2,0\}$. Consumers' purchase decisions are summarized in Fig. 4, where there are five cases.

4.2. Equilibrium outcomes

Similar to Section 3.2, we derive Proposition 2, with the proof provided in Appendix B.5. The expressions for $\dot{\gamma}$ and $\ddot{\gamma}$ are included in the proof of Proposition 2.

Proposition 2. *When consumers are inconsistent, the retailer offers gift card promotions, with the strategies outlined as follows:*

(1) when $\alpha\leq\frac{1}{2}$ *or* $\alpha>\frac{1}{2}$ *and* $\gamma\leq\dot{\gamma}$, *the optimal prices and gift card values are* $p_{1I}^{1*}=\frac{1}{2}$, $p_{2I}^{1*}=1$, *and* $G_I^{1*}=\frac{1}{2}$. *(Strategy 1)*

(2) when $\alpha>\frac{1}{2}$ *and* $\dot{\gamma}<\gamma\leq\ddot{\gamma}$, *the optimal values are* $p_{1I}^{2*}=\frac{1-\alpha(3-\gamma)-2\alpha^2\gamma}{1-2\alpha(3-\gamma)+\alpha^2(1-\gamma)^2}$, $p_{2I}^{2*}=1$, *and* $G_I^{2*}=\frac{2-5\alpha+\alpha^2(1-\gamma)}{1-2\alpha(3-\gamma)+\alpha^2(1-\gamma)^2}$. *(Strategy 2)*

(3) when $\alpha>\frac{1}{2}$ *and* $\gamma>\ddot{\gamma}$, *the optimal strategy involves* $p_{1I}^{3*}=\frac{\gamma-\alpha\gamma(2+\gamma)}{2(1-\gamma)\gamma-2\alpha(1+\gamma-\gamma^2)}$, $p_{2I}^{3*}=1$, *and* $G_I^{3*}=\frac{1-\alpha(2+\gamma)}{2(1-\gamma)\gamma-2\alpha(1+\gamma-\gamma^2)}$. *(Strategy 3)*

Proposition 2 indicates that when consumers are inconsistent, the retailer always implements gift card promotions. Similar to Proposition 1, there are three distinct gift card strategies. Fig. 5 illustrates that the retailer's choice of strategy depends on the proportion of consumers exhibiting double mental discounting and the cash equivalent effect. When both the proportion and effect are relatively low, the retailer employs Strategy 1. As the proportion of double mental discounting consumers rises, the retailer shifts to Strategy 2. When both factors are sufficiently high, the retailer adopts Strategy 3.

Building on Proposition 1, Proposition 2 further demonstrates that consumer inconsistency increases the likelihood of retailers implementing gift card promotions. In this study, consumer inconsistency is characterized by a higher cash equivalent effect during the redemption period than in the

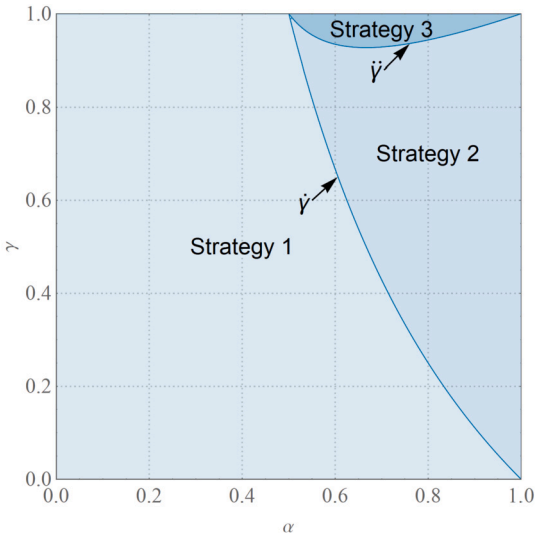


Fig. 5. Retailer's optimal strategies with consumers inconsistency.

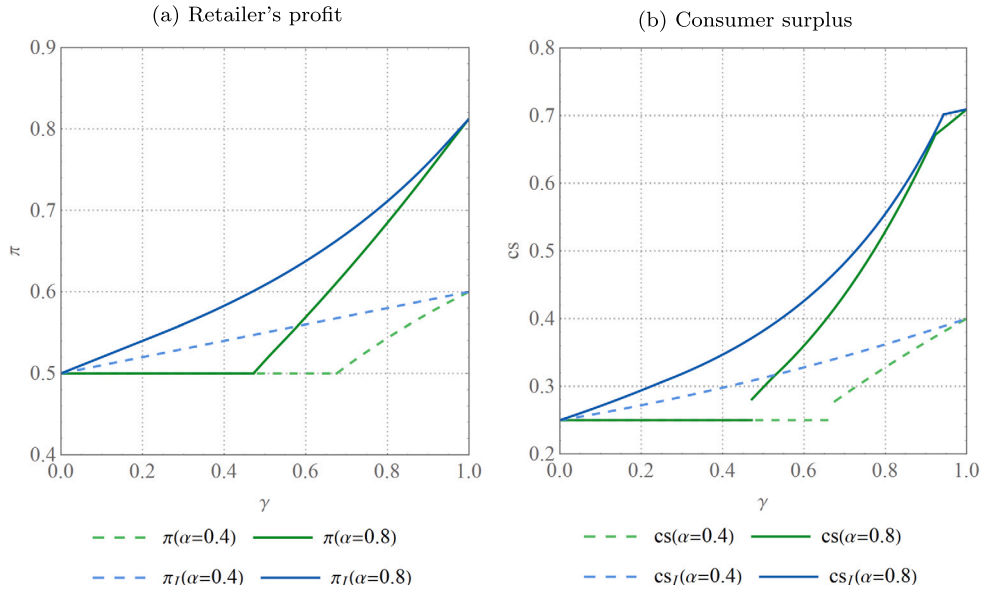


Fig. 6. Impact of cash equivalent effect on profit and consumer surplus.

initial purchase period, which elevates the likelihood of a purchase when the gift card is redeemed. This inconsistency affects the retailer's revenue in two ways: it boosts demand while also incurring redemption costs. Obviously, the increase in revenue exceeds the redemption costs, making gift card promotions profitable for the retailer. Therefore, it is crucial for retailers to consider consumer inconsistency when optimizing their pricing and promotional strategies.

4.3. The impact of consumer inconsistency

Based on Propositions 1 and 2, we evaluate the retailer's profit and consumer surplus under conditions of consumer consistency and inconsistency, as illustrated in Figs. 6 and 7. In these figures, the green line represents consumer consistency, while the blue line represents consumer inconsistency.

Fig. 6 shows that consumer inconsistency increases both retailer profit and consumer surplus. Specifically, Fig. 6a demonstrates that when $\alpha = 0.4$ and $\alpha = 0.6$, the retailer's profit is higher under consumer inconsistency compared to consistency. Moreover, under both consistency and inconsistency, the profit increases with the cash equivalent effect. Fig. 6b indicates that when $\alpha = 0.4$, consumer surplus under inconsistency equals that under consistency when the cash equivalent effect is low but surpasses it when the cash equivalent effect is high. When $\alpha = 0.8$, consumer surplus is consistently higher under inconsistency. Similar to the retailer's profit, consumer surplus also rises with the cash equivalent effect.

Fig. 7 illustrates how the proportion of consumers with double mental discounting affects retailer profit and consumer surplus under conditions of both consumer consistency and inconsistency, with values set at $\gamma = 0.4$ and $\gamma = 0.8$. As shown in Fig. 7a, the retailer's profit is greater under consumer inconsistency compared to consistency, and it increases as the proportion of consumers with double mental discounting rises in both cases. Likewise, Fig. 7b indicates that consumer surplus is higher in scenarios of inconsistency and also grows with the proportion of consumers exhibiting double mental discounting.

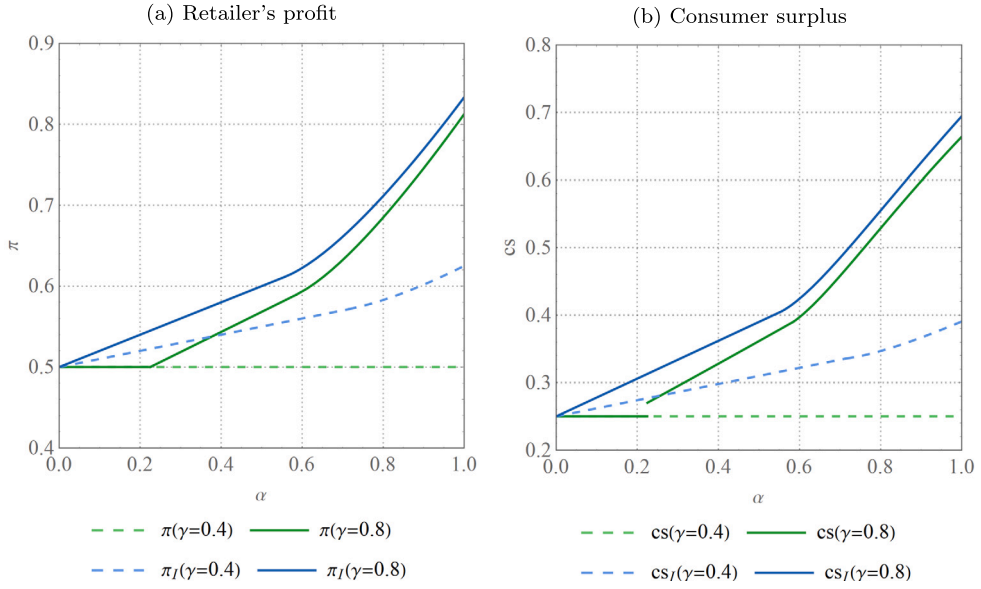


Fig. 7. Impact of double mental discounting effect on profit and consumer surplus.

Figs. 6 and 7 together indicate that consumer inconsistency benefits both retailers and consumers, compared to the consistent consumer scenario. For retailers, the increase in demand from consumer inconsistency more than compensates for redemption costs, enhancing overall profitability. For consumers, this inconsistency results in increased spending during the redemption period while simultaneously lowering their effective purchase cost through gift card deductions. This combined effect results in gains that surpass any additional expenses, ultimately increasing consumer surplus in the inconsistent case.

5. Numerical analysis

The above analysis assumes a uniform distribution of consumers' valuation of products. To test the robustness of the results, we conducted numerical experiments under uniform, exponential and normal distributions, setting $\lambda = 0.02$ for the exponential distribution and $\mu = 0.5$ and $\sigma = 0.1$ for the normal distribution. With a cash equivalent effect of $\gamma = 0.8$, we present the findings in Figs. 8–9.

Fig. 8 illustrates how the proportion of consumers with double mental discounting impacts both profit and gift card value. As shown in Fig. 8a, the retailer's profit increases with the proportion of consumers exhibiting double mental discounting across all three distributions. Notably, the profit is highest under the normal distribution, while the profits under the uniform and exponential distributions are identical. This suggests that understanding consumer valuation accurately is crucial for the retailer to increase revenue. Fig. 8b reveals that gift card values increase with the proportion of consumers with double mental discounting under both exponential and normal distributions, aligning with Corollary 4.

Fig. 9 illustrates how the proportion of consumers with double mental discounting affects prices in the first and second periods. In both Figs. 9a and 9b, product prices increase with a higher proportion of consumers exhibiting double mental discounting. The prices in the uniform and exponential distributions are identical, while the price trends in the normal distribution align with those of the other two distributions, confirming Corollary 3. Additionally, when $\alpha < 0.2$, the retailer does not implement a gift card promotion, and prices remain constant. When $\alpha \geq 0.2$, the retailer initiates gift card promotions, and prices increase, consistent with Corollary 1. By combining Figs. 8 and 9, we observe that the trends in profit, gift card value, and prices are similar across the uniform, exponential, and normal distributions, demonstrating the robustness of the conclusions derived from the uniform distribution.

6. Extensions

In this section, we incorporate consumer heterogeneity in the cash equivalent effect, sales costs, and market competition to validate the robustness of our findings and to derive additional insights.

6.1. Consumers heterogeneity in the cash equivalent effect

Our base model assumes that all consumers share the same cash equivalent effect. Here, we expand the analysis to include heterogeneity in consumers cash equivalent effects. Consumers' valuation of a gift card is influenced not only by objective factors such as store accessibility and product variety but also by subjective factors like gender, age, and income (Gunasti and Baskin, 2018). For instance, Gunasti and Baskin (2018) empirically demonstrate that higher incomes correlate with higher valuations of gift cards. Similarly, a consumer's purchase frequency also affects the perceived value of gift cards. High-frequency shoppers, who view gift cards as directly reducing the cost of future purchases, tend to place a higher value on them. Based on these factors, we assume that consumers' valuation of gift cards can either be high or low. This framework aligns with Mamadehussene (2024), who posit that consumers' redemption costs can also vary between high and low. Specifically, a proportion β of consumers assign a high valuation (i.e., a high cash equivalent effect, γ_H), while the remaining $(1-\beta)$ assign a low valuation (i.e., a low cash equivalent effect, γ_L), with $0 < \gamma_L < \gamma_H$ and $0 < \beta < 1$. Thus, there are four distinct consumer types: those with double mental discounting and a high cash equivalent effect, those with double mental discounting and a low cash equivalent effect, those without double mental discounting and a high cash equivalent effect, and those without double mental discounting and a low cash equivalent effect. For simplicity, we assume $\gamma_H = 1$. Other assumptions regarding

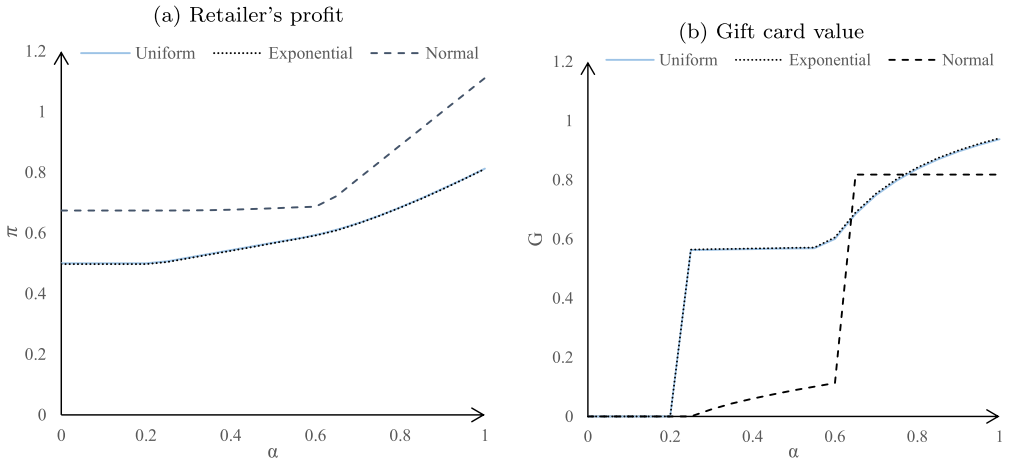


Fig. 8. Effect of double mental discounting on profit and gift card value.

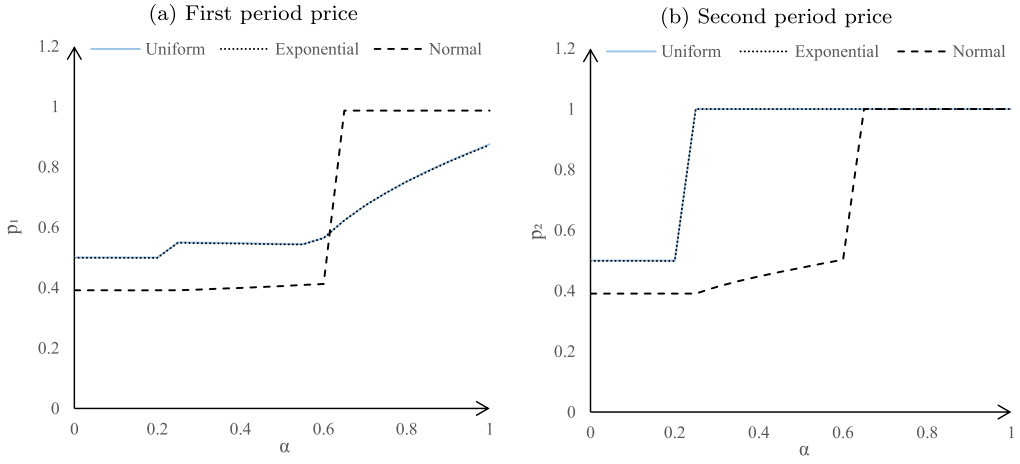


Fig. 9. Effect of double mental discounting on price.

v, p_1, p_2, and G remain consistent with those in Section 3. Proposition 3 confirms that Corollary 2 still holds. The expressions for Ω_1, Ω_2, Ω_3, Ω_4, and Ω_5, along with the detailed proof, are provided in Appendix B.6.

Proposition 3. *Suppose consumers are heterogeneous in cash equivalent effect,*

(1) when $(\beta,\gamma_L,\alpha)\in\Omega_1$, *the retailer does not provide gift card, the optimal prices are* $p_1^{N*}=\frac{1}{2}$ *and* $p_2^{N*}=\frac{1}{2}$. *(No gift card)*

(2) when $(\beta,\gamma_L,\alpha)\in\Omega_2$, *the optimal values are* $p_1^{1*}=\frac{1}{2}$, $p_2^{1*}=1$ *and* $G^{1*}=\frac{1}{2}$. *(Strategy 1)*

(3) when $(\beta,\gamma_L,\alpha)\in\Omega_3$, *the optimal strategy involves* $p_1^{2*}=\frac{1-\beta(2-3\alpha+\gamma_L)+\gamma_L}{2(1+\gamma_L-\alpha\gamma_L-(1-\alpha)\beta(1+\gamma_L))}$, $p_2^{2*}=1$ *and* $G^{2*}=\frac{1-\beta(2-3\alpha+\gamma_L)+\gamma_L}{2(1+\gamma_L-\alpha\gamma_L-(1-\alpha)\beta(1+\gamma_L))}$. *(Strategy 2)*

(4) when $(\beta,\gamma_L,\alpha)\in\Omega_4$, *the retailer sets* $p_1^{3*}=\frac{1}{1+\gamma_L}$, $p_2^{3*}=1$ *and* $G^{3*}=\frac{1}{1+\gamma_L}$. *(Strategy 3)*

(5) when $(\beta,\gamma_L,\alpha)\in\Omega_5$, *the optimal strategy involves* $p_1^{4*}=\frac{\alpha(2+\beta+\gamma_L-\beta\gamma_L)-1}{2\alpha}$, $p_2^{4*}=1$ *and* $G^{4*}=\frac{\alpha(2+\beta+\gamma_L-\beta\gamma_L)-1}{2\alpha}$. *(Strategy 4)*

Proposition 3 demonstrates that, similar to Proposition 1, retailers face the choice of whether to implement a gift card promotion. This decision is influenced by the cash equivalent effect (γ_L), the proportion of consumers with a high cash equivalent effect (β), and the proportion of consumers exhibiting double mental discounting (α). However, in contrast to Proposition 1, four distinct gift card strategies emerge. As shown in Fig. 10, when the proportion of consumers with double mental discounting is relatively low, retailers typically refrain from offering gift card promotions. In contrast, as the proportion of consumers with double mental discounting increases, retailers are more likely to adopt gift card promotions. Depending on the degree of consumer heterogeneity in both double mental discounting and cash equivalent effects, retailers may implement one of the four distinct gift card strategies. Furthermore, we observe that retailers adjust their pricing and promotional strategies in response to consumer heterogeneity in the cash equivalent effect. For instance, in Strategy 1, the first-period price is identical to the price when no gift card is offered. Additionally, the retailer sets the first-period price equal to the value of the gift card when a gift card is provided.

6.2. Sales costs

To simplify our analysis, the main model assumes zero sales costs for retailers. However, in practice, sales costs play a significant role in influencing retailers' decisions. In this subsection, we extend the base model to incorporate sales costs, assuming a unit sales cost of c. All other assumptions

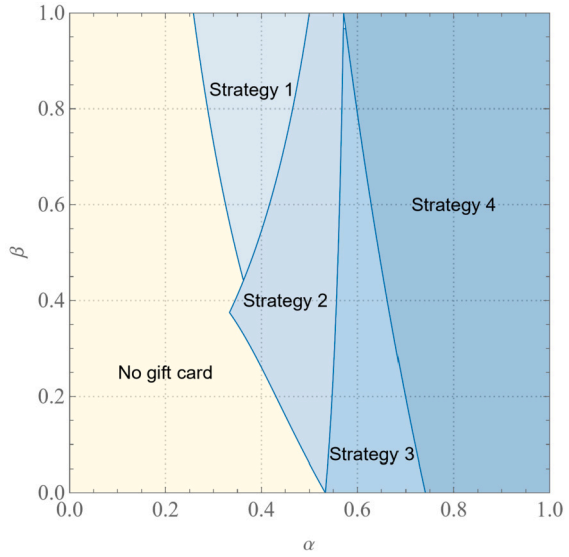


Fig. 10. Optimal strategies with heterogeneous cash equivalent effect ($\gamma_L = 0.6$).



Fig. 11. Retailer's optimal strategies with sale cost ($c = 0.05$).

remain consistent with those in Section 3. The results, shown in Proposition 4, confirm that incorporating sales costs does not alter the key insights of the base model: Corollaries 1(1) and 2 continue to hold. The expression for $\hat{c}$ and detailed proof can be found in Appendix B.7.

Proposition 4. *Suppose the unit cost of selling one product is c,*

(1) when $\gamma \le \hat{\gamma}$, the retailer does not engage in gift card promotions, setting $p_1^{N*} = \frac{1+c}{2}$ and $p_2^{N*} = \frac{1+c}{2}$. (No gift card)

(2) when $\alpha \le \frac{1}{2}$ and $\gamma > \hat{\gamma}$ or $\alpha > \frac{1}{2}$ and $\hat{\gamma} < \gamma \le \tilde{\gamma}$, the retailer offers gift cards with $p_1^{1*} = \frac{2+\alpha\gamma+c(2+\alpha)\gamma}{2(1+\gamma+\alpha\gamma)}$, $p_2^{1*} = 1$, and $G^{1*} = \frac{(1-c)(2+\alpha)}{2(1+\gamma+\alpha\gamma)}$. (Strategy 1)

(3) when $\alpha > \frac{1}{2}$, $\gamma > \tilde{\gamma}$ and $c > \max\{\hat{c}, 0\}$, the retailer sets $p_1^{2*} = \frac{1-2\alpha(1+c\gamma)+\alpha^2(1-2\gamma-(1-2c)\gamma^2)}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$, $p_2^{2*} = 1$, and $G^{2*} = \frac{(1-c)(2+\alpha^2(1-\gamma)-\alpha(3+2\gamma))}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$. (Strategy 2)

(4) when $\alpha > \frac{1}{2}$, $\gamma > \tilde{\gamma}$ and $c \le \hat{c}$, the retailer sets $p_1^{3*} = \frac{1-2c\gamma-\alpha(1+(2-3c)\gamma)}{2-2\alpha(2-\gamma)-2\gamma}$, $p_2^{3*} = 1$, and $G^{3*} = \frac{1-\alpha-2c\gamma-(2-3c)\alpha\gamma}{2\gamma(1-\alpha(2-\gamma)-\gamma)}$. (Strategy 3)

Proposition 4 and Fig. 11 reveal that, similar to Proposition 1 and Fig. 3, the decision for a retailer to implement gift card promotions depends on the proportion of consumers exhibiting double mental discounting (α) and the cash equivalent effect (γ), leading to four distinct strategies. Furthermore, the thresholds for selecting a particular strategy remain consistent with those outlined in Proposition 1, and consumer surplus can be higher when gift cards are offered compared to when they are not, as confirmed by Corollary 2. Sales costs primarily impact the value of both product prices and gift cards. Specifically, sales costs lead the retailer to increase the first-period price while reducing the value of the gift card, which may result in the gift card value being lower than the first-period price.

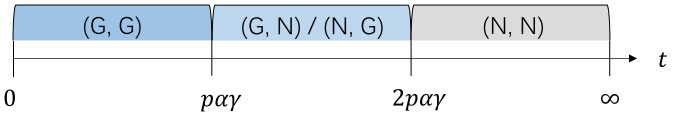


Fig. 12. Competitive retailers' optimal strategies.

6.3. Competition

In this subsection, we expand our analysis to account for competition between two retailers. Specifically, we model two competing firms located at opposite ends of a Hotelling line of unit length—firm a at $x = 0$ and firm b at $x = 1$. Both retailers simultaneously decide whether to offer gift card promotions and, if so, determine the gift card value G_r ($r \in \{a, b\}$) to maximize their profits. Consumers have a base valuation of v for the product, and the product price is denoted as p. We assume that the product price p is exogenously given, which aligns with previous literature (Chen et al., 2022; Yu et al., 2018). All other assumptions are consistent with those outlined in Section 3.

The net utility for a consumer located at position x ($x \in [0, 1]$), when purchasing from firm r ($r = a, b$) during the i-th ($i = 1, 2$) purchase period is given by the following expressions: $U_i^a = v - tx - P_i^a$ and $U_i^b = v - t(1 - x) - P_i^b$, where $P_i^r = p - \gamma G_r$ for consumers with double mental discounting, and $P_1^r = p$ and $P_2^r = p - \gamma G_r$ for consumers not exhibiting this bias. Here, t represents the degree of horizontal differentiation between the two retailers. We assume that v ($v > p + t$) is sufficiently large, ensuring that the market is fully covered. Proposition 5 outlines the optimal strategies of competitive retailers. A detailed proof is provided in Appendix B.8.

Proposition 5. *The optimal strategies for competitive retailers are,*

(1) when $t > 2p\alpha\gamma$, retailers refrain from offering gift card promotions. (N, N)
(2) when $p\alpha\gamma < t \leq 2p\alpha\gamma$, only one retailer offers gift cards, with $G^{1*} = p - \frac{t}{2\alpha\gamma}$. (G, N) or (N, G)
(3) when $t \leq p\alpha\gamma$, both retailers provide gift cards, setting $G_r^{2*} = 2p - \frac{t}{\alpha\gamma}$, $r = a, b$. (G, G)

Proposition 5 indicates that retailers' decisions to implement gift card promotions depend on the degree of horizontal differentiation between the two retailers (t), leading to three distinct strategies. Specifically, Fig. 12 shows that when horizontal differentiation is low, both retailers implement gift card promotions. As horizontal differentiation increases, only one retailer offers a gift card promotion. When horizontal differentiation reaches higher levels, neither retailer provides gift card promotions. Furthermore, the value of the gift card is higher when both retailers participate in the promotion compared to when only one does. This suggests that market competition reduces the purchase cost, benefiting consumers.

7. Conclusions

Gift card promotions are prevalent in retail and often trigger a behavioral phenomenon known as double mental discounting. This bias lowers consumers' perceived prices, which encourages additional spending. While beneficial to retailers, overlooking this effect can lead to substantial potential losses. Most previous research assumes consumers are fully rational, applying the gift card's value as a single deduction upon redemption and overlooking the effects of double mental discounting. This paper fills this gap by analyzing optimal pricing and gift card strategies that incorporate double mental discounting, providing valuable insights for both practitioners and researchers. We begin by analyzing consumer behavior under consistent preferences, then assess the impact of consumer inconsistency, and finally extend our model to incorporate consumer heterogeneity in the cash equivalent effect, retailer sales costs, and market competition.

7.1. Main findings

Our results show that both the double mental discounting and the cash equivalent effects significantly influence the retailer's decision to implement gift card promotions. Specifically, if the cash equivalent effect is low, the best approach is to avoid gift card promotions; otherwise, offering gift cards proves to be profitable. Retailers implement three distinct gift card strategies, with the choice of strategy closely tied to these effects. When offering gift cards, retailers tend to increase prices and offer high-value gift cards to encourage purchases while preserving revenue. The double mental discounting effect positively impacts the perceived value of gift cards but may exert downward pressure on pricing. Under consumer inconsistency, retailers consistently offer gift card promotions, and numerical results indicate that consumer inconsistency enhances both retailer profit and consumer surplus. Additionally, numerical experiments using exponential and normal distributions confirm the robustness of the findings derived from the uniform distribution. Finally, we enhance the basic model by incorporating consumer heterogeneity in cash equivalent effects, retailer sales costs, and market competition, and determine the optimal strategies for firms.

7.2. Theoretical contributions

This research makes several key contributions to the literature. First, it examines the impact of double mental discounting on retailers' decision-making, an area that has received limited attention in prior gift card studies (Chen et al., 2022). Specifically, this paper advances the field by introducing consumer heterogeneity in double mental discounting and incorporating retailers' strategic pricing decisions. This approach offers a more nuanced understanding of how promotional strategies interact with consumer biases and identifies three distinct gift card strategies that retailers may adopt. Second, by considering both the double mental discounting and the cash equivalent effects, we show that retailer decisions are influenced by both effects and that high levels of these effects make offering gift cards profitable. Third, this paper extends mental accounting theory by modeling consumers' double mental discounting in the context of gift card promotions, demonstrating that double mental discounting enhances gift card value and influences pricing strategies. Additionally, our findings provide theoretical support for the empirical results of Cheng and Cryder (2018). Finally, this paper contributes to the broader literature on behavioral biases in pricing and revenue management by highlighting the importance of accounting for double mental discounting in pricing models.

7.3. Managerial implications

Our analysis provides several managerial implications for retailers considering gift card promotions. First, both the proportion of consumers with double mental discounting and the cash equivalent effect play a critical role in determining whether and how to implement gift card promotions. Retailers should pay particular attention to the proportion of consumers with double mental discounting, as a higher proportion can lead to increased consumer spending, benefiting the retailer. Retailers must also account for the cash equivalent effect, which is influenced by factors such as store accessibility and product variety (Gunasti and Baskin, 2018). Adjusting these factors can enhance the cash equivalent effect, making gift card promotions more effective. Second, our model suggests that offering high-value gift cards can be effective, a strategy employed by brands like Apple and some luxury retailers. Double mental discounting lowers consumers' perceived prices, and high-value gift cards further reduce this perception, making it an effective strategy to boost sales and revenue. Third, retailers should carefully assess the proportion of consumers with double mental discounting and the strength of the cash equivalent effect. When this proportion is low but rising, lowering prices may be beneficial, but as it becomes larger, retailers can leverage double mental discounting to raise prices. Similarly, when the cash equivalent effect is low but increasing, lowering prices is advisable, while a stronger effect can justify price increases. Finally, if retailers find that consumers tend to show inconsistency, they may benefit from offering gift card promotions.

For consumers, gift card promotions can enhance surplus. Although double mental discounting reduces their perceived purchase cost and may lead to overspending, Cheng and Cryder (2018) find that consumers are generally aware of the actual costs and are not easily misled by gift card promotions. This duality suggests that the impact of gift cards on consumer surplus can ultimately be positive.

7.4. Limitations and future work

This study primarily examines product-specific gift card promotions, a popular strategy used by brands like Apple. However, purchase-amount-based gift cards, such as tiered incentives where higher spending unlocks higher-value gift cards, are also prevalent (Kadiyala et al., 2024). Our findings may not fully extend to these purchase-amount-based promotions. Tiered gift cards may generate distinct double mental discounting effects due to varying gift card values. Additionally, these promotions often apply to a bundle of products, and the mix of product categories can influence consumers' perceived value of the gift card (Gunasti and Baskin, 2018). Future research could incorporate these factors to deepen the understanding of consumer purchasing behavior and to optimize tiered gift card strategies for retailers.

Several directions for future research merit consideration, including examining the role of different entities in sponsoring gift cards and other promotional forms. In practice, manufacturers sometimes sponsor gift cards (Khouja et al., 2013b). When they do, they must account for retailers' strategic responses, suggesting that future studies could extend this analysis to assess the broader impact on supply chain decisions. Another valuable area for exploration is the comparison between gift cards and other delayed incentives, such as rebates. Rebates tend to trigger a weaker double mental discounting effect than gift cards (Cheng and Cryder, 2018), but they provide cash to consumers without concerns about the perceived value being lower than the face value. When considering both the double mental discounting and cash equivalent effects, it remains unclear whether gift cards are more profitable than rebates. Therefore, future research could investigate the profitability of gift cards versus rebates, offering valuable insights for managers.

CRediT authorship contribution statement

Jie Cui: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jingming Pan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jie Cui reports financial support was provided by National Natural Science Foundation of China. Jingming Pan reports financial support was provided by Natural Science Foundation of Sichuan Province. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Gift card promotion examples

Fig. A.1 illustrates Apple's 2023 Black Friday promotion, where the company offers up to $200 in free gift cards with the purchase of select products. Customers receive an Apple gift card after buying eligible products, which can be used for their next purchase.

In Fig. A.2, Beaverbrooks, a British jewelry brand, provides free gift cards with the purchase of selected items. For instance, buying a luxury watch priced between £1,000 and £1,999 earns the customer a £150 gift card.

Additionally, Fig. A.3 showcases Decathlon, a leading sports equipment retailer, offering free gift cards redeemable for in-store purchases. For every online order of $50, customers receive a $5 gift card, and for every order of $100, they receive a $10 gift card.

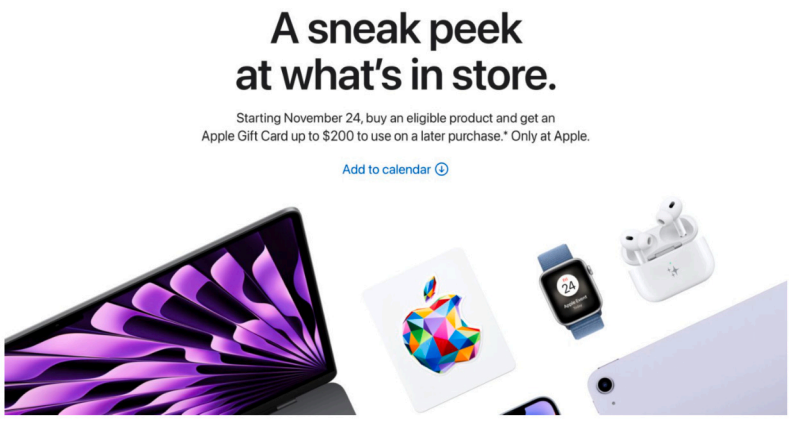


Fig. A.1. Gift card promotions in Apple.

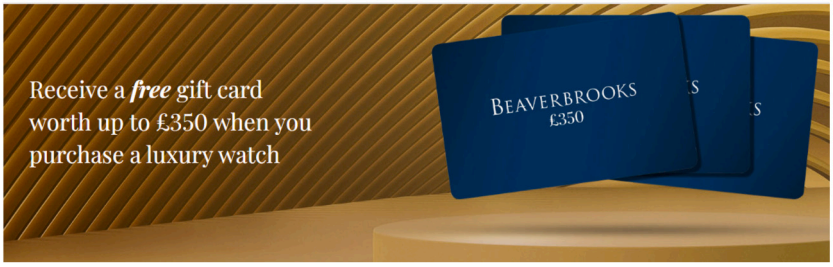


Fig. A.2. Gift card promotions in luxury.

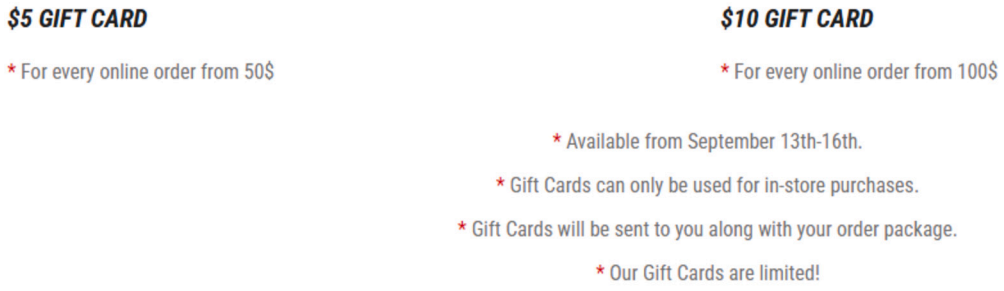


Fig. A.3. Gift card promotions in sports.

Appendix B. Proof

B.1. Proof of Proposition 1, Section 3

Case I Using the Karush-KuhnTucker (KKT) conditions to solve equation (1), we obtain that (1) when $\alpha \le \frac{1}{2}$ or $\alpha > \frac{1}{2}$ and $\gamma \le \frac{1}{2\alpha}$, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$; (2) when $\alpha > \frac{1}{2}$ and $\gamma > \frac{1}{2\alpha}$, $p_1^* = 1$, $p_2^* = 1$, and $G^* = 1$.

Case II Using the KKT conditions to solve equation (2), we obtain the optimal decisions as $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$.

Case III Using the KKT conditions to solve equation (3), we derive that (1) when $\gamma \le \frac{2}{2+2\alpha+\alpha^2}$, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$; (2) when $\alpha \le \frac{1}{2}$ and $\gamma > \frac{2}{2+2\alpha+\alpha^2}$ or $\alpha > \frac{1}{2}$ and $\frac{2}{2+2\alpha+\alpha^2} < \gamma \le \frac{2-\alpha+\alpha^2}{2\alpha+3\alpha^2}$, $p_1^* = \frac{2+\alpha\gamma}{2(1+\gamma+\alpha\gamma)}$, $p_2^* = 1$, and $G^* = \frac{2+\alpha}{2(1+\gamma+\alpha\gamma)}$; (3) when $\alpha > \frac{1}{2}$ and $\frac{2-\alpha+\alpha^2}{2\alpha+3\alpha^2} < \gamma \le \frac{2-3\alpha+3\alpha^2+\sqrt{4-20\alpha+37\alpha^2-26\alpha^3+9\alpha^4}}{4\alpha}$, $p_1^* = \frac{(1-\alpha)^2-\alpha^2\gamma(2+\gamma)}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$, $p_2^* = 1$, and $G^* = \frac{2+\alpha^2(1-\gamma)-\alpha(3+2\gamma)}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$; (4) when $\alpha > \frac{1}{2}$ and $\gamma \ge \frac{2-3\alpha+3\alpha^2+\sqrt{4-20\alpha+37\alpha^2-26\alpha^3+9\alpha^4}}{4\alpha}$, $p_1^* = \frac{1-\alpha-2\alpha\gamma}{2(1-2\alpha-\gamma+\alpha\gamma)}$, $p_2^* = 1$, and $G^* = \frac{1-\alpha-2\alpha\gamma}{2\gamma(1-2\alpha-\gamma+\alpha\gamma)}$.

Case IV Using the KKT conditions to solve equation (4), we obtain that (1) when $\gamma \le \frac{2}{2+2\alpha+\alpha^2}$, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$; (2) when $\gamma > \frac{2}{2+2\alpha+\alpha^2}$, $p_1^* = \frac{2+\alpha\gamma}{2(1+\gamma+\alpha\gamma)}$, $p_2^* = 1$, and $G^* = \frac{2+\alpha}{2(1+\gamma+\alpha\gamma)}$.

Define $\hat{\gamma} = \frac{2}{2+2\alpha+\alpha^2}$, $\tilde{\gamma} = \frac{2-\alpha+\alpha^2}{2\alpha+3\alpha^2}$, and $\bar{\gamma} = \frac{2-3\alpha+3\alpha^2+\sqrt{4-20\alpha+37\alpha^2-26\alpha^3+9\alpha^4}}{4\alpha}$. By combining these cases, we drive Proposition 1.

B.2. Proof of Corollary 1, Section 3

By comparing optimal decisions with and without gift cards in Proposition 1, we find that (1) $p_1^{j*} > p_1^{N*}$ and $p_2^{j*} > p_1^{N*}$, $j = 1, 2, 3$. (2) $G^{j*} > p_1^{j*}$, $j = 1, 2, 3$.

B.3. Proof of Corollary 2, Section 3

According Proposition 1, we obtain consumer surplus: $cs^N = \frac{1}{4}$, $cs^1 = \frac{(2+\alpha)^2(2+3\alpha)\gamma^2}{8(1+\gamma+\alpha\gamma)^2}$, $cs^2 = \frac{\alpha\gamma^2(4-4\alpha(3+2\gamma)+\alpha^4(1-3\gamma^2)+2\alpha^3(-4+\gamma+2\gamma^2)+\alpha^2(17+8\gamma+4\gamma^2))}{(1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma))^2}$, and $cs^3 = \frac{2(1-2\gamma)^2-\alpha(9-40\gamma+36\gamma^2)-\alpha^3(1-36\gamma+24\gamma^2)+4\alpha^2(3-17\gamma+14\gamma^2)}{8(1-\alpha(2-\gamma)-\gamma)^2}$.

Define $\hat{\alpha}$ is the valid real solution of equation $1-24\alpha+96\alpha^2-116\alpha^3+8\alpha^4+8\alpha^5=0$, $\acute{\gamma}$ is the valid real solution of equation $1-4\alpha+6\alpha^2-4\alpha^3+\alpha^4-(4\alpha-4\alpha^2+4\alpha^3+4\alpha^4)\gamma-(16\alpha-54\alpha^2+64\alpha^3-38\alpha^4+4\alpha^5)\gamma^2+(32\alpha^2-36\alpha^3-12\alpha^4)\gamma^3-(16\alpha^3+15\alpha^4-12\alpha^5)\gamma^4=0$, $\grave{\gamma} = \frac{\sqrt{2}(-1+7\alpha-15\alpha^2+9\alpha^3)+\sqrt{2-25\alpha+130\alpha^2-362\alpha^3+566\alpha^4-465\alpha^5+150\alpha^6}}{\sqrt{2}(-3+16\alpha-27\alpha^2+12\alpha^3)}$, and $\check{\gamma} = \frac{\sqrt{2}(-1+7\alpha-15\alpha^2+9\alpha^3)-\sqrt{2-25\alpha+130\alpha^2-362\alpha^3+566\alpha^4-465\alpha^5+150\alpha^6}}{\sqrt{2}(-3+16\alpha-27\alpha^2+12\alpha^3)}$.

By comparing, we get that (1) when $\gamma \le \frac{2(1+\alpha)+\sqrt{2(8+20\alpha+14\alpha^2+3\alpha^3)}}{6+16\alpha+12\alpha^2+3\alpha^3}$, $cs^1 \le cs^N$; otherwise, $cs^1 > cs^N$. (2) When $\alpha \le \hat{\alpha}$, or $\alpha > \hat{\alpha}$ and $\gamma \le \acute{\gamma}$, $cs^2 \le cs^N$, otherwise, $cs^2 > cs^N$. (3) When $\alpha \le 2-\sqrt{3}$ and $\gamma \le \grave{\gamma}$, or $2-\sqrt{3} < \alpha \le \frac{9-\sqrt{33}}{12}$ and $\check{\gamma} < \gamma \le \grave{\gamma}$, $cs^3 \le cs^N$, otherwise, $cs^3 > cs^N$.

B.4. Proof of Corollaries 3–6, Section 3

Corollary 3

By taking the derivative of p_1^{j*} for $j=1,2,3$ with respect to α, we get that $\frac{\partial p_1^{1*}}{\partial\alpha} = -\frac{(1-\gamma)\gamma}{2(1+\gamma+\alpha\gamma)^2} < 0$, $\frac{\partial p_1^{2*}}{\partial\alpha} = \frac{2\gamma(1-2\alpha\gamma+\alpha^2(4\gamma+\gamma^2-1))}{(1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma))^2} > 0$, and $\frac{\partial p_1^{3*}}{\partial\alpha} = \frac{1-2\gamma+2\gamma^2}{2(1-\alpha(2-\gamma)-\gamma)^2} > 0$.

Corollary 4

By taking the derivative of G^{j*}, $j=1,2,3$ with respect to α, we get that $\frac{\partial G^{1*}}{\partial\alpha} = \frac{1-\gamma}{2(1+\gamma+\alpha\gamma)^2} > 0$, $\frac{\partial G^{2*}}{\partial\alpha} = \frac{1+2\gamma+\alpha(-2+6\gamma-4\gamma^2)+\alpha^2(1-4\gamma+\gamma^2+2\gamma^3)}{(1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma))^2} > 0$, and $\frac{\partial G^{3*}}{\partial\alpha} = \frac{1-2\gamma+2\gamma^2}{2(1-\alpha(2-\gamma)-\gamma)^2\gamma} > 0$.

Corollary 5

By taking the derivative of p_1^{j*} for $j=1,2,3$ with respect to γ, we get that $\frac{\partial p_1^{1*}}{\partial\gamma} = -\frac{2+\alpha}{2(1+\gamma+\alpha\gamma)^2} < 0$, $\frac{\partial p_1^{2*}}{\partial\gamma} = \frac{2\alpha(1-2\alpha^3(1-\gamma)\gamma-2\alpha(1+\gamma)+\alpha^2(1+4\gamma+\gamma^2))}{(1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma))^2} > 0$, and $\frac{\partial p_1^{3*}}{\partial\gamma} = \frac{1-4\alpha+5\alpha^2}{2(1-\alpha(2-\gamma)-\gamma)^2} > 0$.

Corollary 6

By taking the derivative of G^{j*} for $j=1,2,3$ with respect to γ, we get that $\frac{\partial G^{1*}}{\partial\gamma} = -\frac{(1+\alpha)(2+\alpha)}{2(1+\gamma+\alpha\gamma)^2} < 0$, $\frac{\partial G^{2*}}{\partial\gamma} = \frac{\alpha(2+\alpha+\alpha^3(1-\gamma)^2-4\alpha\gamma+2\alpha^2(-2+3\gamma+\gamma^2))}{(1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma))^2} > 0$, and when $\alpha < 1/2$ and $\gamma < \frac{1-\alpha-\sqrt{1-4\alpha+5\alpha^2}}{2\alpha}$ or $\alpha > 1/2$ and $\gamma > \frac{1-\alpha+\sqrt{1-4\alpha+5\alpha^2}}{2\alpha}$, $\frac{\partial G^{3*}}{\partial\gamma} = -\frac{(1-\alpha)(1-2\gamma+2\alpha(\gamma+\gamma^2-1))}{2(1-\alpha(2-\gamma)-\gamma)^2\gamma^2} < 0$, otherwise, $\frac{\partial G^{3*}}{\partial\gamma} > 0$.

B.5. Proof of Proposition 2, Section 4

Case I and II

$$\max_{p_1,p_2,G} \pi = p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + p_2(1-\alpha)\int_{p_2}^{p_1} 1\,dx + (p_2-G)\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right), \quad \text{subject to} \quad 1 \ge p_1 \ge p_2 > p_1-\gamma G \ge 0 \quad \text{and} \quad 1 \ge G \ge 0. \tag{B.1}$$

$$\max_{p_1,p_2,G} \pi = (p_1+p_2-G)\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + p_2\left(\alpha\int_{p_2}^{p_1-\gamma G} 1\,dx + (1-\alpha)\int_{p_2}^{p_1} 1\,dx\right), \quad \text{subject to} \quad 1 \ge p_1 > p_1-\gamma G \ge p_2 \ge 0 \quad \text{and} \quad 1 \ge G \ge 0. \tag{B.2}$$

Case I and II are same as consumers consistency, we are not repeating it here.

Case III

$$\max_{p_1,p_2,G} \pi = p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + (p_2-G)\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right), \quad \text{subject to} \quad 1 > p_2 \ge p_1 > p_1-\gamma G > p_2-G \ge 0 \quad \text{and} \quad 1 \ge G \ge 0. \tag{B.3}$$

Using the KKT conditions to solve equation (B.3), we obtain that (1) when $\alpha \le \frac{1}{2}$, $p_1^* = \frac{1}{2}$, and $p_2^* = \frac{1}{2}$; (2) when $\alpha > \frac{1}{2}$, $p_1^* = \frac{1}{2-\gamma}$, $p_2^* = 1$, and $G^* = \frac{1}{2-\gamma}$.

Case IV

$$\max_{p_1,p_2,G} \pi = p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + (p_2-G)\left(\alpha\int_{p_2-G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right), \quad \text{subject to} \quad 1 > p_2 \ge p_1 > p_2-G > p_1-\gamma G \ge 0 \quad \text{and} \quad 1 \ge G \ge 0. \tag{B.4}$$

Using the KKT conditions to solve equation (B.4), we derive that (1) when $\alpha \le \frac{1}{2}$ or $\alpha > \frac{1}{2}$ and $\gamma \le \frac{1-\alpha}{\alpha}$, $p_1^* = \frac{1}{2}$, $p_2^* = 1$, and $G^* = \frac{1}{2}$; (2) when $\alpha > \frac{1}{2}$ and $\frac{1-\alpha}{\alpha} < \gamma \le \frac{2-6\alpha+3\alpha^2+\sqrt{4-24\alpha+44\alpha^2-24\alpha^3+9\alpha^4}}{2\alpha^2}$, $p_1^* = \frac{1-\alpha(3-\gamma)-2\alpha^2\gamma}{1+2\alpha(-3+\gamma)+\alpha^2(1-\gamma)^2}$, $p_2^* = 1$, and $G^* = \frac{2-5\alpha+\alpha^2(1-\gamma)}{1+2\alpha(-3+\gamma)+\alpha^2(1-\gamma)^2}$; (3) when $\alpha > \frac{1}{2}$ and $\gamma > \frac{2-6\alpha+3\alpha^2+\sqrt{4-24\alpha+44\alpha^2-24\alpha^3+9\alpha^4}}{2\alpha^2}$, $p_1^* = \frac{\gamma-\alpha\gamma(2+\gamma)}{2(1-\gamma)\gamma-2\alpha(1+\gamma-\gamma^2)}$, $p_2^* = 1$, and $G^* = \frac{1-\alpha(2+\gamma)}{2(1-\gamma)\gamma-2\alpha(1+\gamma-\gamma^2)}$.

Case V

$$\max_{p_1,p_2,G} \pi = p_1\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + (p_2-G)\left(\alpha\int_{p_2-G}^{1} 1\,dx + (1-\alpha)\int_{p_2-G}^{1} 1\,dx\right),$$
$$\text{subject to} \quad 1 > p_2 > p_2 - G \geq p_1 > p_1 - \gamma G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.5}$$

By applying the KKT conditions to equation (B.5), we derive that $p_1^* = \frac{1}{2}$, $p_2^* = 1$, and $G^* = \frac{1}{2}$.

Define $\dot{\gamma} = \frac{1-\alpha}{\alpha}$ and $\ddot{\gamma} = \frac{2-6\alpha+3\alpha^2+\sqrt{4-24\alpha+44\alpha^2-24\alpha^3+9\alpha^4}}{2\alpha^2}$. By combining the insights from the five distinct cases considered in the analysis, Proposition 2 is formulated.

B.6. Proof of Proposition 3, Section 6

Case I

$$\max_{p_1,p_2,G} \pi = (p_1+p_2-G)\left(\alpha\beta\int_{p_1-\gamma_H G}^{1} 1\,dx + \alpha(1-\beta)\int_{p_1-\gamma_L G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + p_2((1-\alpha)\int_{p_2}^{p_1} 1\,dx),$$
$$\text{subject to} \quad 1 \geq p_1 \geq p_2 > p_1 - \gamma_L G \geq p_1 - \gamma_H G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.6}$$

Using the Karush-KuhnTucker (KKT) conditions to solve equation (B.6), we obtain that (1) when $\gamma_L \leq \frac{1}{2}$ and $\alpha \leq \min\{\frac{1}{2(\beta+\gamma_L-\beta\gamma_L)}, 1\}$, or $\gamma_L > \frac{1}{2}$ and $\alpha \leq \frac{1}{2(\beta+\gamma_L-\beta\gamma_L)}$, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$; (2) when $\gamma_L \leq \frac{1}{2}$, $\beta > \frac{1-2\gamma_L}{2-2\gamma_L}$ and $\alpha > \frac{1}{2(\beta+\gamma_L-\beta\gamma_L)}$, or $\gamma_L > \frac{1}{2}$ and $\alpha > \frac{1}{2(\beta+\gamma_L-\beta\gamma_L)}$, $p_1^* = 1$, $p_2^* = 1$, and $G^* = 1$.

Case II

$$\max_{p_1,p_2,G} \pi = (p_1+p_2-G)\left(\alpha\beta\int_{p_1-\gamma_H G}^{1} 1\,dx + \alpha(1-\beta)\int_{p_1-\gamma_L G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + p_2(\alpha(1-\beta)\int_{p_2}^{p_1-\gamma_L G} 1\,dx + (1-\alpha)\int_{p_2}^{p_1} 1\,dx),$$
$$\text{subject to} \quad 1 \geq p_1 > p_1 - \gamma_L G \geq p_2 \geq p_1 - \gamma_H G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.7}$$

Define $\bar{\alpha} = \frac{(\beta(1-\gamma_L)+\gamma_L)(1-\gamma_L)}{(2\beta(1-\gamma_L)+\gamma_L)^2} + \sqrt{\frac{\gamma_L^2(1+2(-1+\gamma_L)\gamma_L)+\beta^2(1-\gamma_L)^2(1-\gamma_L(2-5\gamma_L))+2\beta\gamma_L(1-\gamma_L(3-(5-3\gamma_L)\gamma_L))}{(2\beta(1-\gamma_L)+\gamma_L)^4}}$. Using the KKT conditions to solve equation (B.7), we obtain that (1) when $\beta \leq \frac{1}{2}$, or $\beta > \frac{1}{2}$ and $\alpha \leq \bar{\alpha}$, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$; (2) when $\beta > \frac{1}{2}$ and $\alpha > \bar{\alpha}$, $p_1^* = 1$, and $p_2^* = \frac{2\alpha\beta(1-\gamma_L)+\alpha(2-\gamma_L)\gamma_L+\gamma_L^2}{2\alpha\gamma_L+2\gamma_L^2+2\alpha\beta\left(1-\gamma_L^2\right)}$, $G^* = \frac{\gamma_L+\alpha(2\beta(1-\gamma_L)+\gamma_L)}{2\left(\gamma_L^2+\alpha\left(\beta+\gamma_L-\beta\gamma_L^2\right)\right)}$.

Case III

$$\max_{p_1,p_2,G} \pi = (p_1+p_2-G)\left(\alpha\beta\int_{p_1-\gamma_H G}^{1} 1\,dx + \alpha(1-\beta)\int_{p_1-\gamma_L G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + p_2(\alpha\beta\int_{p_2}^{p_1-\gamma_H G} 1\,dx + \alpha(1-\beta)\int_{p_2}^{p_1-\gamma_L G} 1\,dx + (1-\alpha)\int_{p_2}^{p_1} 1\,dx),$$
$$\text{subject to} \quad 1 \geq p_1 > p_1 - \gamma_L G > p_1 - \gamma_H G > p_2 \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.8}$$

Using the KKT conditions to solve equation (B.8), we obtain the optimal decisions are $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$.

Case IV

$$\max_{p_1,p_2,G} \pi = p_1\left(\alpha\beta\int_{p_1-\gamma_H G}^{1} 1\,dx + \alpha(1-\beta)\int_{p_1-\gamma_L G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + (p_2-G)\left(\beta\int_{p_2-\gamma_H G}^{1} 1\,dx + (1-\beta)\int_{p_2-\gamma_L G}^{1} 1\,dx\right),$$
$$\text{subject to} \quad 1 \geq p_2 > p_2 - \gamma_L G > p_2 - \gamma_H G > p_1 \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.9}$$

Using the KKT conditions to solve equation (B.9), we derive that (1) when $\gamma_L \leq \frac{1}{2}$ and $\alpha \leq \min\{\frac{(1-\beta)(1-\gamma_L)}{\beta+\gamma_L-\beta\gamma_L}, 1\}$, or $\gamma_L > \frac{1}{2}$ and $\alpha \leq \frac{(1-\beta)(1-\gamma_L)}{\beta+\gamma_L-\beta\gamma_L}$, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$; (2) when $\gamma_L \leq \frac{1}{2}$, $\beta > \frac{1-2\gamma_L}{2-2\gamma_L}$ and $\alpha > \frac{(1-\beta)(1-\gamma_L)}{\beta+\gamma_L-\beta\gamma_L}$, or $\gamma_L > \frac{1}{2}$ and $\alpha > \frac{(1-\beta)(1-\gamma_L)}{\beta+\gamma_L-\beta\gamma_L}$, $p_1^* = \frac{1}{2}$, $p_2^* = 1$, and $G^* = \frac{1}{2}$.

Case V

$$\max_{p_1,p_2,G} \pi = p_1\left(\alpha\beta\int_{p_1-\gamma_H G}^{1} 1\,dx + \alpha(1-\beta)\int_{p_1-\gamma_L G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + (p_2-G)\left(\alpha\beta\int_{p_2-\gamma_H G}^{1} 1\,dx + (1-\alpha)\beta\int_{p_1}^{1} 1\,dx + (1-\beta)\int_{p_2-\gamma_L G}^{1} 1\,dx\right),$$
$$\text{subject to} \quad 1 \geq p_2 > p_2 - \gamma_L G > p_1 > p_2 - \gamma_H G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.10}$$

Define $\tilde{\beta}$ is the second valid solution of equation $2-2\beta-7\beta^2+8\beta^3 = 0$, $\hat{\gamma}_L = \frac{\sqrt{2}-1-\beta}{1-\beta}$, $\tilde{\gamma}_L = \frac{1-4\beta+\sqrt{1+4\beta^2}}{2(1-\beta)}$, $\alpha_1 = \frac{\beta^2(2-\gamma_L)-\gamma_L+2\beta\gamma_L}{\beta(5\beta+4(1-\beta)\gamma_L)} + \sqrt{\frac{\beta^2(5+2\beta(2\beta-5))+4(1-\beta)\beta(1+(\beta-3)\beta)\gamma_L-(1-\beta)^2(2\beta+4\beta^2-1)\gamma_L^2+4(\beta-1)^3\beta\gamma_L^3}{\beta^2(\beta(5-4\gamma_L)+4\gamma_L)^2}}$, $\alpha_2 = \frac{\beta}{2\beta+\gamma_L-\beta\gamma_L}$, $\alpha_4 = \frac{1+\beta\gamma_L-\gamma_L^2+\beta\gamma_L^2}{\beta+2\gamma_L+\beta\gamma_L}$, $\dot{\gamma}_L = \frac{\sqrt{2-2\beta+\beta^2}-1}{1-\beta}$, and $\dot{\alpha} = \frac{1-\beta-\gamma_L+\beta\gamma_L}{\beta+\gamma_L-\beta\gamma_L}$.

Using the KKT conditions to solve equation (B.10), we derive that (1) when $\beta \leq \sqrt{2}-1$, $\hat{\gamma}_L < \gamma_L \leq \tilde{\gamma}_L$ and $\alpha_1 < \alpha \leq \min\{\alpha_4, 1\}$ or $\gamma_L > \tilde{\gamma}_L$ and $\alpha_2 < \alpha \leq \alpha_4$, or $\sqrt{2}-1 < \beta \leq \tilde{\beta}$, $\gamma_L \leq \tilde{\gamma}_L$ and $\alpha_1 < \alpha \leq \min\{\alpha_4, 1\}$ or $\gamma_L > \tilde{\gamma}_L$ and $\alpha_2 < \alpha \leq \alpha_4$, or $\tilde{\beta} < \beta \leq \frac{2}{3}$, $\gamma_L \leq \tilde{\gamma}_L$ and $\alpha > \alpha_1$ or $\tilde{\gamma}_L < \gamma_L \leq \dot{\gamma}_L$ and $\alpha > \alpha_2$ or $\gamma_L > \dot{\gamma}_L$ and $\alpha_2 < \alpha < \alpha_4$, or $\beta > \frac{2}{3}$, $\gamma_L \leq \dot{\gamma}_L$ and $\alpha > \alpha_2$ or $\gamma_L > \dot{\gamma}_L$ and $\alpha_2 < \alpha \leq \alpha_4$, $p_1^* = \frac{1-\beta(2-3\alpha+\gamma_L)+\gamma_L}{2(1+\gamma_L-\alpha\gamma_L-(1-\alpha)\beta(1+\gamma_L))}$, $p_2^* = 1$, and $G^* = \frac{1-\beta(2-3\alpha+\gamma_L)+\gamma_L}{2(1+\gamma_L-\alpha\gamma_L-(1-\alpha)\beta(1+\gamma_L))}$; (2) when $\beta \leq \frac{2}{3}$, $\gamma_L > \tilde{\gamma}_L$, and $\dot{\alpha} < \alpha \leq \alpha_2$, or $\beta > \frac{2}{3}$ and $\dot{\alpha} < \alpha \leq \alpha_2$, $p_1^* = \frac{1}{2}$, $p_2^* = 1$, and $G^* = \frac{1}{2}$; (3) when $\gamma_L > \dot{\gamma}_L$ and $\alpha > \alpha_4$, $p_1^* = \frac{1}{1+\gamma_L}$, $p_2^* = 1$, and $G^* = \frac{1}{1+\gamma_L}$; (4) otherwise, $p_1^* = \frac{1}{2}$, $p_2^* = \frac{1}{2}$, and $G^* = 0$.

Case VI

$$\max_{p_1,p_2,G}\pi = p_1\left(\alpha\beta\int_{p_1-\gamma_H G}^{1}1\,dx+\alpha(1-\beta)\int_{p_1-\gamma_L G}^{1}1\,dx+(1-\alpha)\int_{p_1}^{1}1\,dx\right)+(p_2-G)\left(\alpha\beta\int_{p_2-\gamma_H G}^{1}1\,dx+\alpha(1-\beta)\int_{p_2-\gamma_L G}^{1}1\,dx+(1-\alpha)\int_{p_1}^{1}1\,dx\right),$$
$$\text{subject to}\quad 1\geq p_2>p_1>p_2-\gamma_L G>p_2-\gamma_H G\geq 0\quad\text{and}\quad 1\geq G\geq 0. \tag{B.11}$$

Using the KKT conditions to solve equation (B.11), we derive that (1) when $\gamma_L\leq\sqrt{2}-1$ and $\alpha\leq\min\{\frac{1-\gamma_L^2}{2\beta+2\gamma_L-2\beta\gamma_L^2},1\}$, or $\gamma_L>\sqrt{2}-1$ and $\alpha\leq\frac{1-\gamma_L^2}{2\beta+2\gamma_L-2\beta\gamma_L^2}$, $p_1^*=\frac{1}{2}$, $p_2^*=\frac{1}{2}$, and $G^*=0$; (2) when $\gamma_L\leq\sqrt{2}-1$, $\beta>\frac{1-2\gamma_L-\gamma_L^2}{2-2\gamma_L^2}$ and $\frac{1-\gamma_L^2}{2\beta+2\gamma_L-2\beta\gamma_L^2}<\alpha\leq\min\{\frac{1+\gamma_L}{\beta+3\gamma_L+\gamma_L^2-\beta\gamma_L^2},1\}$, or $\gamma_L>\sqrt{2}-1$ and $\frac{1-\gamma_L^2}{2\beta+2\gamma_L-2\beta\gamma_L^2}<\alpha\leq\frac{1+\gamma_L}{\beta+3\gamma_L+\gamma_L^2-\beta\gamma_L^2}$, $p_1^*=\frac{1}{1+\gamma_L}$, $p_2^*=1$, and $G^*=\frac{1}{1+\gamma_L}$; (3) when $\beta>\max\{\frac{1-2\gamma_L-\gamma_L^2}{1-\gamma_L^2},0\}$ and $\alpha>\frac{1+\gamma_L}{\beta+3\gamma_L+\gamma_L^2-\beta\gamma_L^2}$, $p_1^*=\frac{\alpha(2+\beta+\gamma_L-\beta\gamma_L)-1}{2\alpha}$, $p_2^*=1$, and $G^*=\frac{\alpha(2+\beta+\gamma_L-\beta\gamma_L)-1}{2\alpha}$.

Let $\hat{\beta}$ be the first valid solution of equation $9\beta-19\beta^2+43\beta^3-76\beta^4+84\beta^5-48\beta^6+16\beta^7-1=0$, $\bar{\gamma}_L$ is the third valid solution of equation $2\beta-2\beta^2+(1-3\beta)\gamma_L-(2\beta-2\beta^2)\gamma_L^2-(1-\beta)\gamma_L^3=0$, $\hat{\gamma}_L$ is the first valid solution of equation $2\beta-2\beta^2+(1-3\beta)\gamma_L-(2\beta-2\beta^2)\gamma_L^2-(1-\beta)\gamma_L^3=0$, $\alpha_3=\frac{1-\gamma_L^2}{2\beta+2\gamma_L-2\beta\gamma_L^2}$, and $\alpha_5=\frac{1+\gamma_L}{\beta+3\gamma_L+\gamma_L^2-\beta\gamma_L^2}$. We also define the following regions in the β-γ_L-α space: $\Omega_1=\{(\beta,\gamma_L,\alpha)|(\beta\leq\hat{\beta},((\gamma_L\leq\hat{\gamma}_L,0<\alpha<1)\cup(\hat{\gamma}_L<\gamma_L\leq\tilde{\gamma}_L,\alpha\leq\alpha_1)\cup(\tilde{\gamma}_L<\gamma_L\leq\bar{\gamma}_L,\alpha\leq\alpha_2)\cup(\gamma_L>\bar{\gamma}_L,\alpha\leq\alpha_3)))\cup(\hat{\beta}<\beta\leq\sqrt{2}-1,((\gamma_L\leq\hat{\gamma}_L,0<\alpha<1)\cup(\hat{\gamma}_L<\gamma_L\leq\tilde{\gamma}_L,\alpha\leq\alpha_1)\cup(\tilde{\gamma}_L<\gamma_L\leq\hat{\gamma}_L,\alpha\leq\alpha_2)\cup(\gamma_L>\hat{\gamma}_L,\alpha\leq\alpha_3)))\cup(\sqrt{2}-1<\beta\leq\frac{2}{3},((\gamma_L\leq\tilde{\gamma}_L,\alpha\leq\alpha_1)\cup(\tilde{\gamma}_L<\gamma_L\leq\hat{\gamma}_L,\alpha\leq\alpha_2)\cup(\gamma_L>\hat{\gamma}_L,\alpha\leq\alpha_3)))\cup(\beta>\frac{2}{3},((\gamma_L\leq\hat{\gamma}_L,\alpha\leq\alpha_2)\cup(\gamma_L>\hat{\gamma}_L,\alpha\leq\alpha_3)))\}$, $\Omega_2=\{(\beta,\gamma_L,\alpha)|(\beta\leq\hat{\beta},\gamma_L>\bar{\gamma}_L,\alpha_3<\alpha\leq\alpha_2)\cup(\beta>\hat{\beta},\gamma_L>\hat{\gamma}_L,\alpha_3<\alpha\leq\alpha_2)\}$, $\Omega_3=\{(\beta,\gamma_L,\alpha)|(\beta\leq\sqrt{2}-1,((\hat{\gamma}_L<\gamma_L\leq\hat{\gamma}_L,\alpha>\alpha_1)\cup(\hat{\gamma}_L<\gamma_L\leq\tilde{\gamma}_L,\alpha_1<\alpha\leq\alpha_4)\cup(\gamma_L>\tilde{\gamma}_L,\alpha_2<\alpha\leq\alpha_4)))\cup(\sqrt{2}-1<\beta\leq\tilde{\beta},((\gamma_L\leq\hat{\gamma}_L,\alpha>\alpha_1)\cup(\hat{\gamma}_L<\gamma_L\leq\tilde{\gamma},\alpha_1<\alpha\leq\alpha_4)\cup(\gamma_L>\tilde{\gamma},\alpha_2<\alpha\leq\alpha_4)))\cup(\tilde{\beta}<\beta\leq\frac{2}{3},((\gamma_L\leq\tilde{\gamma}_L,\alpha>\alpha_1)\cup(\tilde{\gamma}_L<\gamma_L\leq\hat{\gamma}_L,\alpha>\alpha_2)\cup(\gamma_L>\hat{\gamma}_L,\alpha_2<\alpha\leq\alpha_4)))\cup(\beta>\frac{2}{3},((\gamma_L\leq\hat{\gamma}_L,\alpha>\alpha_2)\cup(\gamma_L>\hat{\gamma}_L,\alpha_2<\alpha\leq\alpha_4)))\}$, $\Omega_4=\{(\beta,\gamma_L,\alpha)|0<\beta<1,\gamma_L>\hat{\gamma}_L,\alpha_4<\alpha\leq\alpha_5\}$, and $\Omega_5=\{(\beta,\gamma_L,\alpha)|0<\beta<1,\gamma_L>\hat{\gamma}_L,\alpha>\alpha_5\}$.

By combining these cases, we drive Proposition 3.

By comparing optimal decisions with and without gift cards in Proposition 3, we find that (1) $p_1^{1*}=p_1^{N*}$, $p_1^{j*}>p_1^{N*}$, $j=2,3,4$, and $p_2^{j*}>p_1^{N*}$, $j=1,2,3,4$. (2) $G^{j*}=p_1^{j*}$, $j=1,2,3,4$.

According Proposition 3, we obtain consumer surplus: $cs^N=\frac{1}{4}$, $cs^1=\frac{1}{8}\left(1+\gamma_L^2+\alpha\gamma_L\left(2+\gamma_L\right)+\beta\left(1-\gamma_L\right)\left(1+\gamma_L+\alpha\left(3+\gamma_L\right)\right)\right)$, $cs^2=\frac{1}{8(1+\gamma_L-\alpha\gamma_L+(-1+\alpha)\beta(1+\gamma_L))^2}\left(1+\beta+\alpha\beta-14\alpha\beta^2+19\alpha^2\beta^2+8\alpha\beta^3-19\alpha^2\beta^3+11\alpha^3\beta^3-2(1-\beta)\left(\alpha-1-\left(1-2\alpha+4\alpha^2\right)\beta+\alpha\left(7-\alpha\left(17-11\alpha\right)\right)\beta^2\right)\gamma_L+(1-\beta)\left(2+\alpha-(4+\alpha(3-2\alpha(11-8\alpha)))\beta+(3-\alpha(3+\alpha(19-25\alpha)))\beta^2\right)\gamma_L^2+2(1-\beta)^2(1-2\beta+\alpha(2-\alpha(2-5\beta)))\gamma_L^3-(1+\alpha)(\beta-1)^3\gamma_L^4\right)$, $cs^3=\frac{2\gamma_L^2+3\alpha\gamma_L^2+\alpha\beta\left(2+2\gamma_L-4\gamma_L^2\right)}{2(1+\gamma_L)^2}$, and $cs^4=\frac{1}{8\alpha^2}\left(2+\alpha\left(-1-2\left(3-\gamma_L\right)\gamma_L-2\alpha^2\beta^3\left(\gamma_L-1\right)^3\gamma_L+\alpha\beta^2\left(1-\gamma_L\right)^2\left(2-4\gamma_L+\alpha\left(3+6\gamma_L\left(1+\gamma_L\right)\right)\right)\right)+2\beta\left(1-\gamma_L\right)\left(\gamma_L-2+\alpha\left(-1-4\gamma_L^2+\alpha\left(4+3\gamma_L\left(1+\gamma_L\right)^2\right)\right)\right)+\alpha\gamma_L\left(6+\gamma_L\left(\alpha\left(3+2\gamma_L\left(3+\gamma_L\right)\right)-2-4\gamma_L\right)\right)\right)$.

By comparing, we obtain that under certain condition, $cs^j>cs^N$, $j=1,2,3,4$.

B.7. Proof of Proposition 4, Section 6

Case I

$$\max_{p_1,p_2,G}\pi=(p_1-c+p_2-c-G)\left(\alpha\int_{p_1-\gamma G}^{1}1\,dx+(1-\alpha)\int_{p_1}^{1}1\,dx\right)+(p_2-c)(1-\alpha)\int_{p_2}^{p_1}1\,dx,$$
$$\text{subject to}\quad 1\geq p_1\geq p_2>p_1-\gamma G\geq 0\quad\text{and}\quad 1\geq G\geq 0. \tag{B.12}$$

Using the Karush-KuhnTucker (KKT) conditions to solve equation (B.12), we obtain that (1) when $\alpha\leq\frac{1}{2}$ or $\alpha>\frac{1}{2}$ and $\gamma\leq\frac{1}{2\alpha}$, $p_1^*=\frac{1+c}{2}$, $p_2^*=\frac{1+c}{2}$, and $G^*=0$; (2) when $\alpha>\frac{1}{2}$ and $\gamma>\frac{1}{2\alpha}$, $p_1^*=1$, $p_2^*=1$, and $G^*=1-c$.

Case II

$$\max_{p_1,p_2,G}\pi=(p_1-c+p_2-c-G)\left(\alpha\int_{p_1-\gamma G}^{1}1\,dx+(1-\alpha)\int_{p_1}^{1}1\,dx\right)+(p_2-c)\left(\alpha\int_{p_2}^{p_1-\gamma G}1\,dx+(1-\alpha)\int_{p_2}^{p_1}1\,dx\right),$$
$$\text{subject to}\quad 1\geq p_1>p_1-\gamma G\geq p_2\geq 0\quad\text{and}\quad 1\geq G\geq 0. \tag{B.13}$$

Using the KKT conditions to solve equation (B.13), we obtain the optimal decisions as $p_1^*=\frac{1+c}{2}$, $p_2^*=\frac{1+c}{2}$, and $G^*=0$.

Case III

$$\max_{p_1,p_2,G}\pi=(p_1-c)\left(\alpha\int_{p_1-\gamma G}^{1}1\,dx+(1-\alpha)\int_{p_1}^{1}1\,dx\right)+(p_2-c-G)\left(\alpha\int_{p_2-\gamma G}^{1}1\,dx+(1-\alpha)\int_{p_1}^{1}1\,dx\right),$$
$$\text{subject to}\quad 1\geq p_2\geq p_1>p_2-\gamma G>p_1-\gamma G\geq 0\quad\text{and}\quad 1\geq G\geq 0. \tag{B.14}$$

Let $\hat{c}=\frac{-1+2\alpha-\alpha^2+2\gamma-3\alpha\gamma+3\alpha^2\gamma-2\alpha\gamma^2}{2\gamma-5\alpha\gamma+\alpha^2\gamma-2\alpha\gamma^2+\alpha^2\gamma^2}$. Using the KKT conditions to solve equation (B.14), we derive that (1) when $\gamma\leq\hat{\gamma}$, $p_1^*=\frac{1+c}{2}$, $p_2^*=\frac{1+c}{2}$, and $G^*=0$; (2) when $\alpha\leq\frac{1}{2}$ and $\gamma>\hat{\gamma}$ or $\alpha>\frac{1}{2}$ and $\hat{\gamma}<\gamma\leq\tilde{\gamma}$, $p_1^*=\frac{2+\alpha\gamma+c(2+\alpha)\gamma}{2(1+\gamma+\alpha\gamma)}$, $p_2^*=1$, and $G^*=\frac{(1-c)(2+\alpha)}{2(1+\gamma+\alpha\gamma)}$; (3) when $\alpha>\frac{1}{2}$, $\gamma>\tilde{\gamma}$ and $c>\max\{\hat{c},0\}$, $p_1^*=\frac{1-2\alpha(1+c\gamma)+\alpha^2(1-2\gamma-(1-2c)\gamma^2)}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$, $p_2^*=1$, and $G^*=\frac{(1-c)(2+\alpha^2(1-\gamma)-\alpha(3+2\gamma))}{1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)}$; (4) when $\alpha>\frac{1}{2}$, $\gamma\geq\bar{\gamma}$ and $c\leq\hat{c}$, $p_1^*=\frac{1-2c\gamma-\alpha(1+(2-3c)\gamma)}{2-2\alpha(2-\gamma)-2\gamma}$, $p_2^*=1$, and $G^*=\frac{1-\alpha-2c\gamma-(2-3c)\alpha\gamma}{2\gamma(1-\alpha(2-\gamma)-\gamma)}$.

Table B.2
Consumers' demand.

Case	Consumer type	Retailer a		Retailer b	
		Period 1	Period 2	Period 1	Period 2
Case I	α	$\frac{t+\gamma(G_a-G_b)}{2t}$	$\frac{t+\gamma(G_a-G_b)}{2t}$	$1-\frac{t+\gamma(G_a-G_b)}{2t}$	$1-\frac{t+\gamma(G_a-G_b)}{2t}$
	$1-\alpha$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
Case II	α	$\frac{t+\gamma G_a}{2t}$	$\frac{t+\gamma G_a}{2t}$	$1-\frac{t+\gamma G_a}{2t}$	$1-\frac{t+\gamma G_a}{2t}$
	$1-\alpha$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
Case III		$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

Case IV

$$\max_{p_1,p_2,G} \pi = (p_1-c)\left(\alpha\int_{p_1-\gamma G}^{1} 1\,dx + (1-\alpha)\int_{p_1}^{1} 1\,dx\right) + (p_2-c-G)\int_{p_2-\gamma G}^{1} 1\,dx, \quad \text{subject to} \quad 1 \geq p_2 > p_2-\gamma G \geq p_1 > p_1-\gamma G \geq 0 \quad \text{and} \quad 1 \geq G \geq 0. \tag{B.15}$$

Using the KKT conditions to solve equation (B.15), we obtain that (1) when $\gamma \leq \hat{\gamma}$, $p_1^* = \frac{1+c}{2}$, $p_2^* = \frac{1+c}{2}$, and $G^* = 0$; (2) when $\gamma > \hat{\gamma}$, $p_1^* = \frac{2+\alpha\gamma+c(2+\alpha)\gamma}{2(1+\gamma+\alpha\gamma)}$, $p_2^* = 1$, and $G^* = \frac{(1-c)(2+\alpha)}{2(1+\gamma+\alpha\gamma)}$.

By combining these cases, we drive Proposition 4.

By comparing optimal decisions with and without gift cards in Proposition 4, we find that (1) $p_1^{j*} > p_1^{N*}$ and $p_2^{j*} > p_1^{N*}$, $j = 1,2,3$. (2) When $c \leq \frac{\alpha-\alpha\gamma}{2+\alpha+2\gamma+\alpha\gamma}$, $G^{1*} \geq p_1^{1*}$; otherwise, $G^{1*} < p_1^{1*}$. When $c \leq \frac{1-\alpha-2\alpha\gamma+\alpha^2\gamma+\alpha^2\gamma^2}{2-3\alpha+\alpha^2-4\alpha\gamma-\alpha^2\gamma+2\alpha^2\gamma^2}$, $G^{2*} \geq p_1^{2*}$; otherwise, $G^{2*} < p_1^{2*}$. $G^{3*} > p_1^{3*}$.

According Proposition 4, we obtain consumer surplus: $cs^N = \frac{(1-c)^2}{4}$, $cs^1 = \frac{(1-c)^2(2+\alpha)^2(2+3\alpha)\gamma^2}{8(1+\gamma+\alpha\gamma)^2}$,
$cs^2 = \frac{(1-c)^2\alpha\gamma^2\left(4-4\alpha(3+2\gamma)-2\alpha^3\left(4-\gamma-2\gamma^2\right)+\alpha^4\left(1-3\gamma^2\right)+\alpha^2\left(17+8\gamma+4\gamma^2\right)\right)}{\left(1+\alpha^2(1-\gamma)^2-2\alpha(1+\gamma)\right)^2}$, and $cs^3 = \left(2\left(1-2(1-c)\gamma\right)^2 - \alpha^3\left(1-6(6-7c)\gamma+3\left(8-12c+3c^2\right)\gamma^2\right) - \alpha\big(9-8(5-6c)\gamma+4\left(9-18c+7c^2\right)\gamma^2\right)+2\alpha^2\left(6-(34-41c)\gamma+\left(28-48c+15c^2\right)\gamma^2\right)\big)/(8(1-\alpha(2-\gamma)-\gamma)^2)$.

By comparing, we obtain that under certain condition, $cs^j > cs^N$, $j = 1,2,3$.

B.8. Proof of Proposition 5, Section 6

There are three possible cases: (1) both retailers offer a gift card promotion (Case I), (2) only one retailer offers a gift card promotion (Case II), and (3) neither retailer offers a gift card promotion (Case III).

Consumers' demand in each case is summarized in Table B.2.

Case I

$$\begin{cases} \pi_a = (2p-G_a)\left(\alpha\dfrac{t+\gamma(G_a-G_b)}{2t}+\dfrac{1-\alpha}{2}\right) \\ \pi_b = (2p-G_b)\left(\alpha\left(1-\dfrac{t+\gamma(G_a-G_b)}{2t}\right)+\dfrac{1-\alpha}{2}\right) \end{cases} \tag{B.16}$$

Since $\frac{\partial^2\pi_a}{\partial G_a^2} < 0$ and $\frac{\partial^2\pi_b}{\partial G_b^2} < 0$, both π_a and π_b are concave in G_a and G_b, respectively. From the first-order-condition, the optimal gift card values are given by $\hat{G}_a = \hat{G}_b = 2p - \frac{t}{\alpha\gamma}$. Upon further analysis, we obtain the following results: (1) when $t \geq 2p\alpha\gamma$, $\hat{G}_k \leq 0$ $(k = a,b)$, which implies $G_a^* = G_b^* = 0$, and neither retailer offers a gift card promotion; (2) when $t < 2p\alpha\gamma$, $\hat{G}_k > 0$ $(k = a,b)$, and the optimal gift card values are $G_a^* = G_b^* = 2p - \frac{t}{\alpha\gamma}$, meaning both retailers implement gift card promotions.

Case II

$$\begin{cases} \pi_a = (2p-G_a)\left(\alpha\dfrac{t+\gamma G_a}{2t}+\dfrac{(1-\alpha)}{2}\right) \\ \pi_b = 2p\left(\alpha\left(1-\dfrac{t+\gamma G_a}{2t}\right)+\dfrac{(1-\alpha)}{2}\right) \end{cases} \tag{B.17}$$

Since $\frac{\partial^2\pi_a}{\partial G_a^2} < 0$, it follows that π_a is concave in G_a. From the first-order-condition, the optimal gift card values are given by $\tilde{G}_a = p - \frac{t}{2\alpha\gamma}$. Through analysis, we obtain that (1) when $t \geq 2p\alpha\gamma$, $\tilde{G}_a \leq 0$, implying that $G_a^* = 0$. In this case, no retailer offers a gift card promotion; (2) when $t < 2p\alpha\gamma$, $\tilde{G}_a > 0$, and the optimal gift card value is $G_a^* = p - \frac{t}{2\alpha\gamma}$. In this case, one retailer provides a gift card promotion.

Case III

When no retailer provides gift card, $\pi_a = \pi_b = p$.

By comparing these cases, we drive Proposition 5.

Data availability

No data was used for the research described in the article.

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